

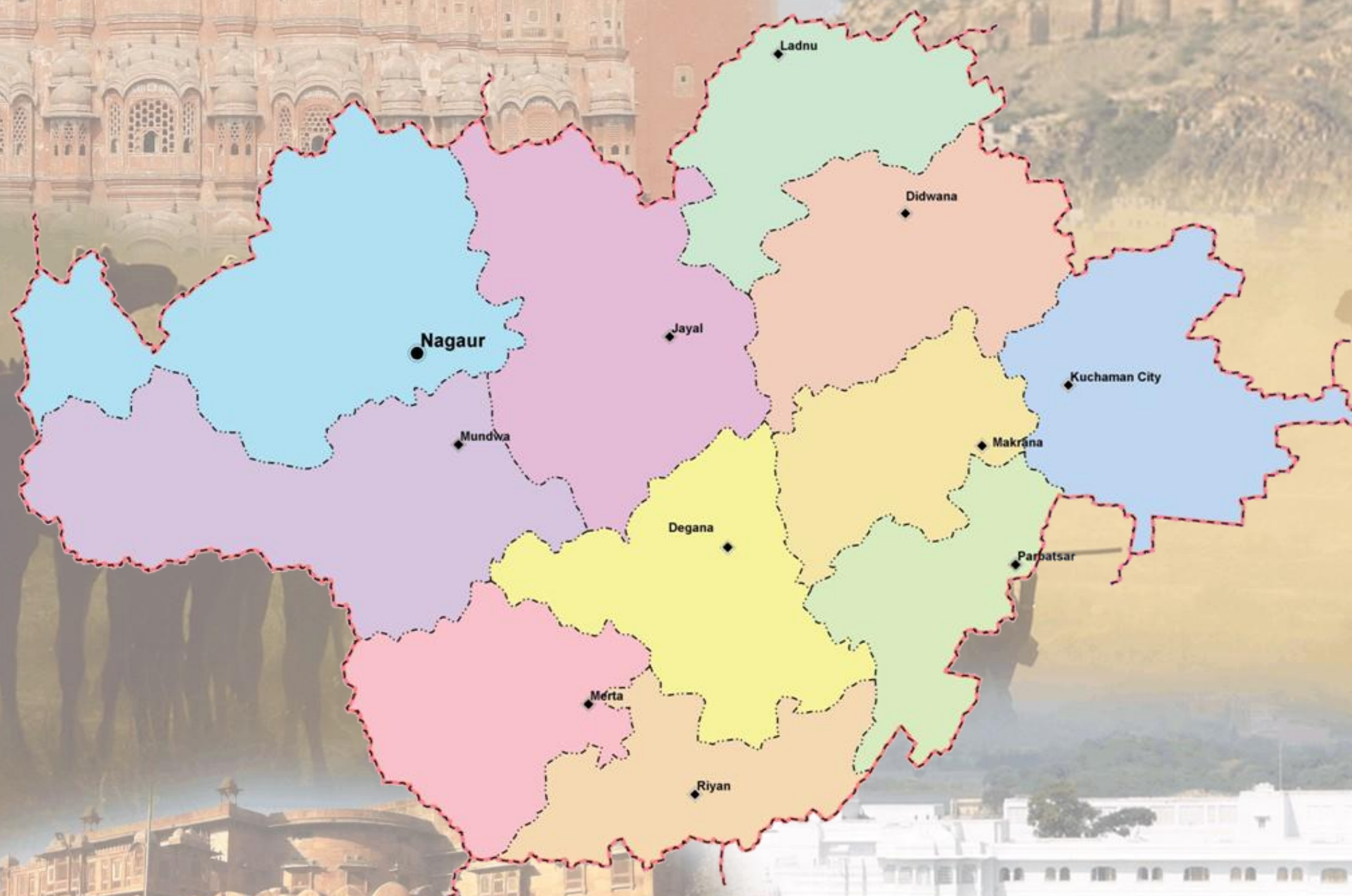


Ground Water Department,  
Rajasthan

# Hydrogeological Atlas of Rajasthan Nagaur District



European Union  
State Partnership Programme



2013



Rolta India Limited

## Message



**Shri Ashok Gehlot,**  
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

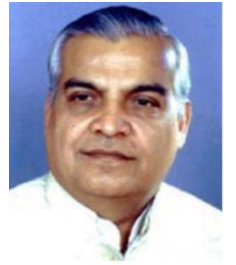
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

**(Ashok Gehlot)**

## Message



**Dr. Jitendra Singh,**  
Hon'ble Minister for Energy,  
Non-Conventional Energy Resources,  
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

**(Jitendra Singh)**

## Message



**Dr. Purushottam Agarwal**

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

**(Purushottam Agarwal)**

## Foreword



**Arno Schaefer**

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

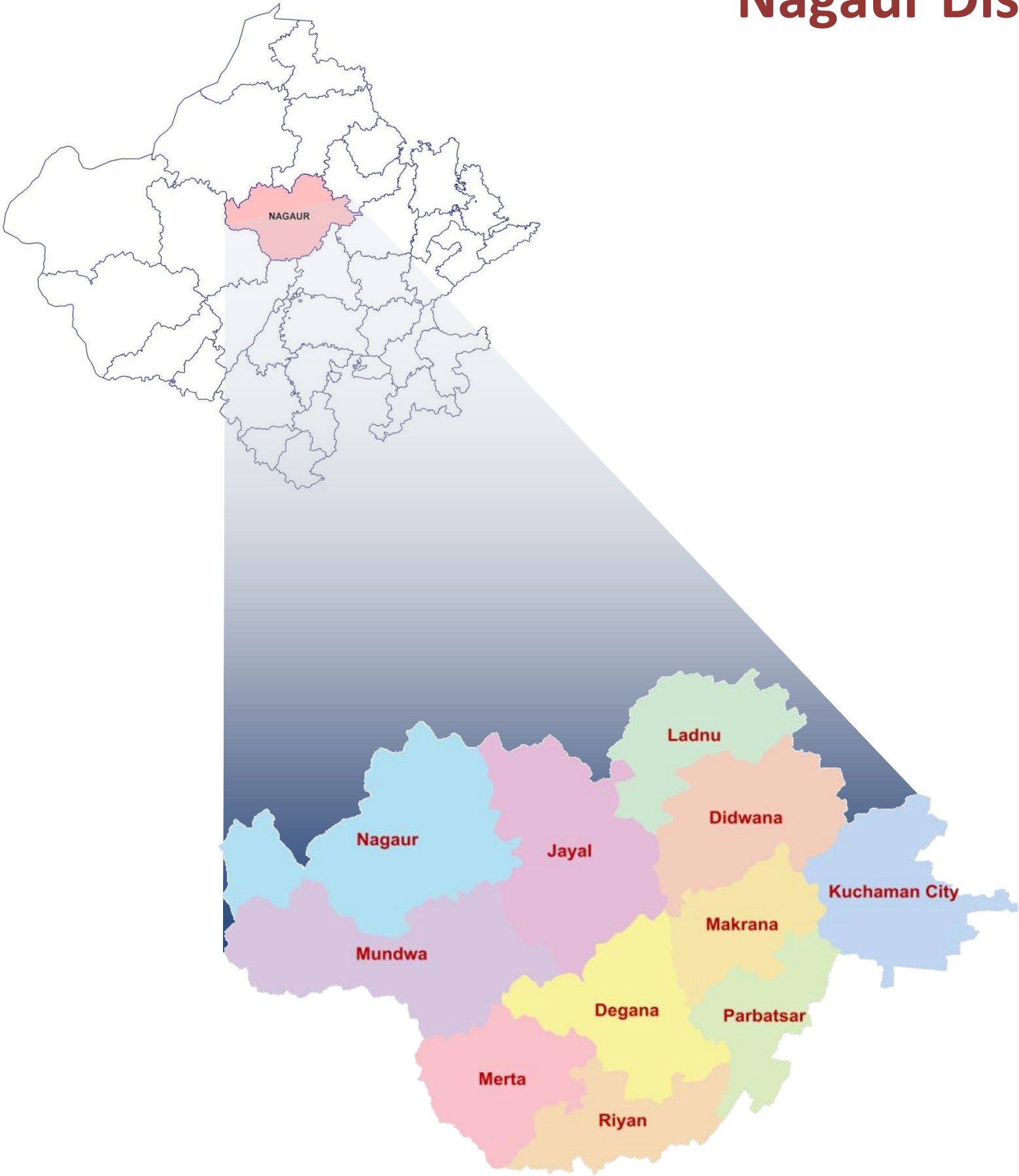
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

**(Arno Schaefer)**

# Hydrogeological Atlas of Rajasthan

## Nagaur District



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## Location:

Nagaur district is located in the central part of Rajasthan. It is bounded in the north by Bikaner and Churu districts, in the east by Sikar and Jaipur districts, south by Pali and Ajmer districts and the western boundary is bounded by Jodhpur district. It stretches between 26° 24' 01.10" to 27° 42' 28.21" north latitude and 73° 04' 43.46" to 75° 22' 04.08" east longitude covering area of 17,805.1 sq km. Major part of the district does not have a systematic drainage system, so most of the central and northern region is part of 'Outside' Basin. The Luni River drains in the southwestern part and Sekhawati River drains in the southeastern part of Nagaur district.

## Administrative Set-up:

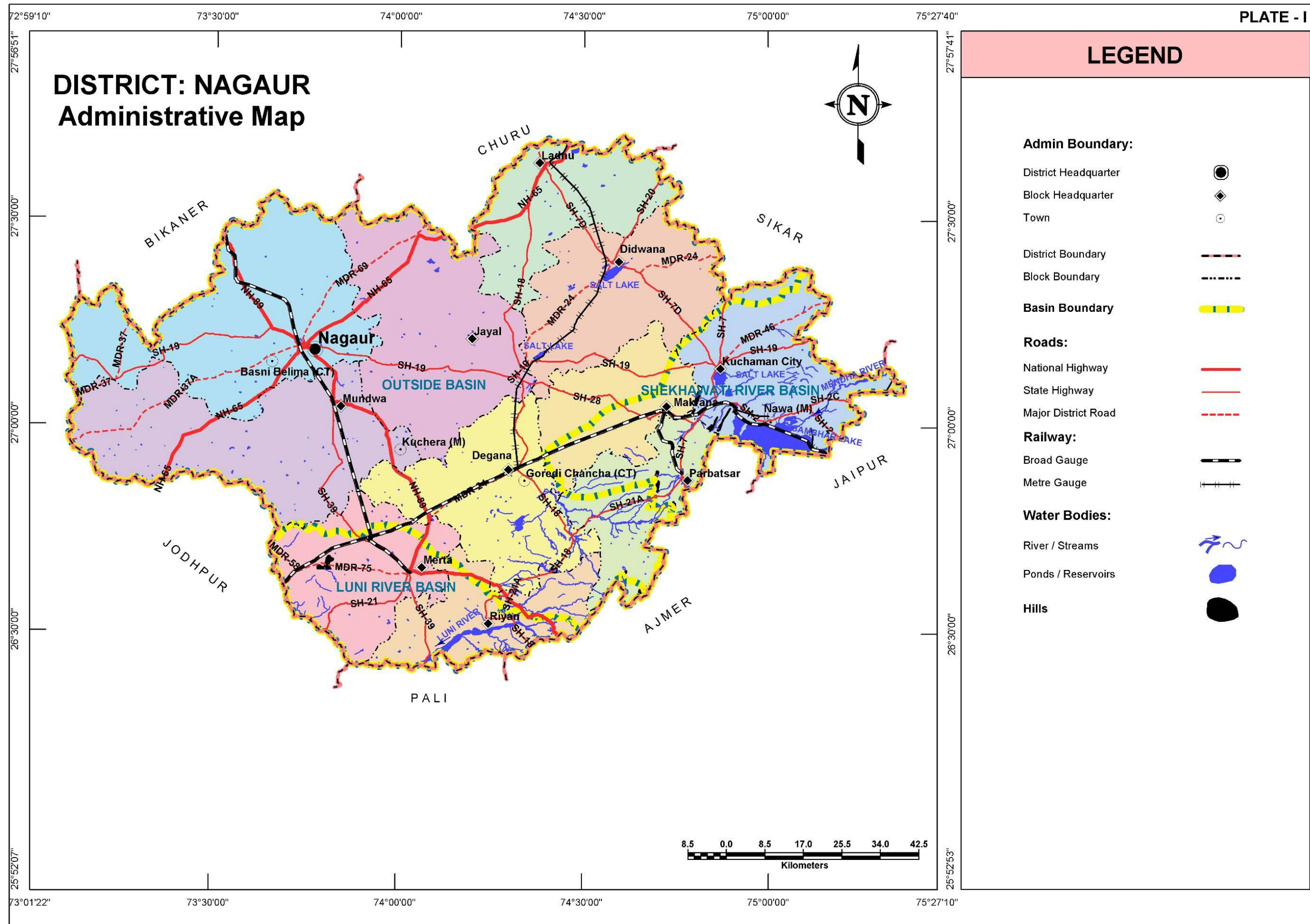
Nagaur district is administratively divided into eleven blocks. The following table summarizes the basic statistics of the district at block level.

| S. No.       | Block Name    | Population<br>(Based on 2001 census) | Area<br>(sq km) | % of District<br>Area | Total Number of<br>Towns and Villages |
|--------------|---------------|--------------------------------------|-----------------|-----------------------|---------------------------------------|
| 1            | Degana        | 1,99,923                             | 1,578.7         | 9.0                   | 153                                   |
| 2            | Didwana       | 3,29,706                             | 1,641.9         | 9.2                   | 173                                   |
| 3            | Jayal         | 2,24,944                             | 2,201.2         | 12.3                  | 137                                   |
| 4            | Kuchaman City | 3,36,963                             | 1,632.1         | 9.1                   | 195                                   |
| 5            | Ladnu         | 2,15,096                             | 1,275.3         | 7.0                   | 99                                    |
| 6            | Makrana       | 2,83,000                             | 1,076.3         | 6.0                   | 120                                   |
| 7            | Merta         | 2,23,681                             | 1,356.8         | 8.0                   | 116                                   |
| 8            | Mundwa        | 2,33,070                             | 2,335.4         | 13.1                  | 124                                   |
| 9            | Nagaur        | 3,35,201                             | 2,555.0         | 14.3                  | 156                                   |
| 10           | Parbatsar     | 2,00,234                             | 1,045.6         | 6.0                   | 111                                   |
| 11           | Riyan         | 1,79,629                             | 1,106.8         | 6.0                   | 128                                   |
| <b>Total</b> |               | <b>27,61,447</b>                     | <b>17,805.1</b> | <b>100.0</b>          | <b>1,512</b>                          |

Nagaur district has 1,512 towns and villages, of which eleven are block headquarters as well.

## Climate:

Nagaur is largely part of arid climate with hot summer. Sand storms are common in summer. The climate of the district is marked by extreme dryness, large variations of temperature & highly irregular rainfall patterns. The maximum temperature recorded in district is 47.2 °C with 0 °C as the lowest recorded temperature. The average temperature of the district is 23.3°C. The winter season extends from mid-November until the beginning of March. The rainy season is relatively short, extending from July through mid-September. The average rainfall in the district is 349.8mm.



## TOPOGRAPHY

## DISTRICT – NAGAUR

The general topography of the district is undulating. Eastern part of the district comprises small scattered hillocks. The slope of the area is towards west. The northwestern part of the district is extensively covered with sand dunes. The general topography in the district ranges between 275 m to 375 m above mean sea level. Elevation ranges from a minimum of 275.0 m above mean sea level in Nagaur block in the NW part of the district and maximum of 794.0 m above mean sea level in Parbatsar block in eastern part of the district.

**Table: Block wise minimum and maximum elevation**

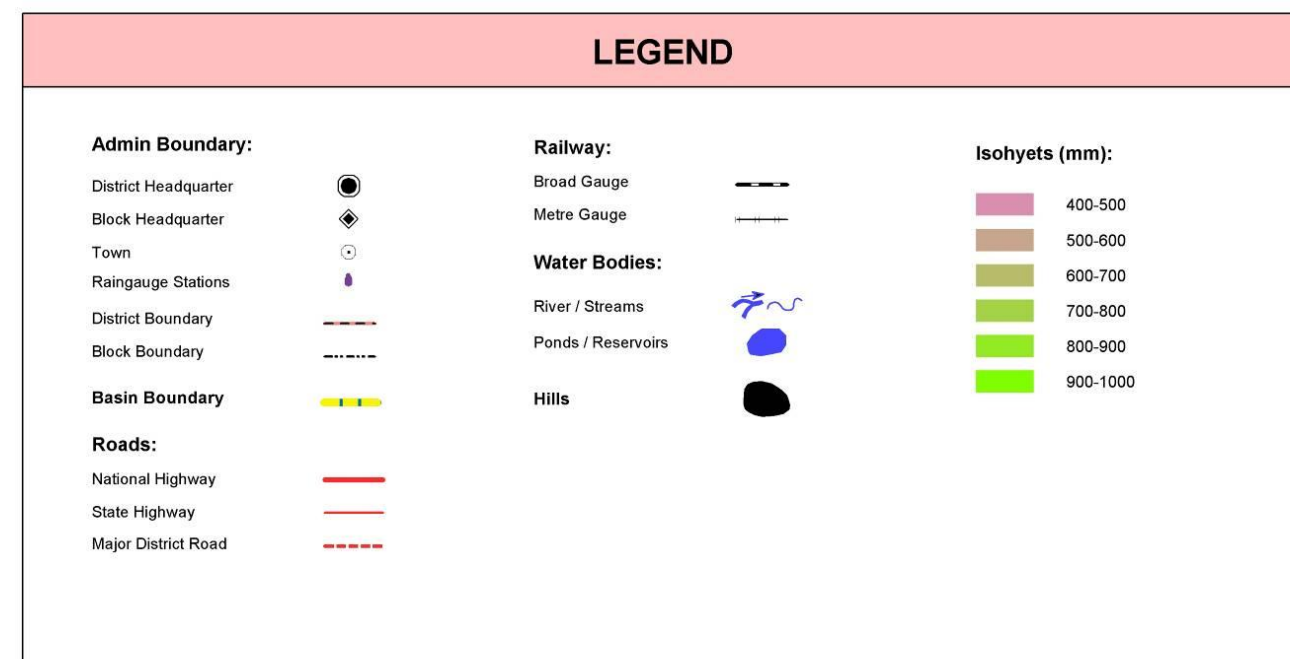
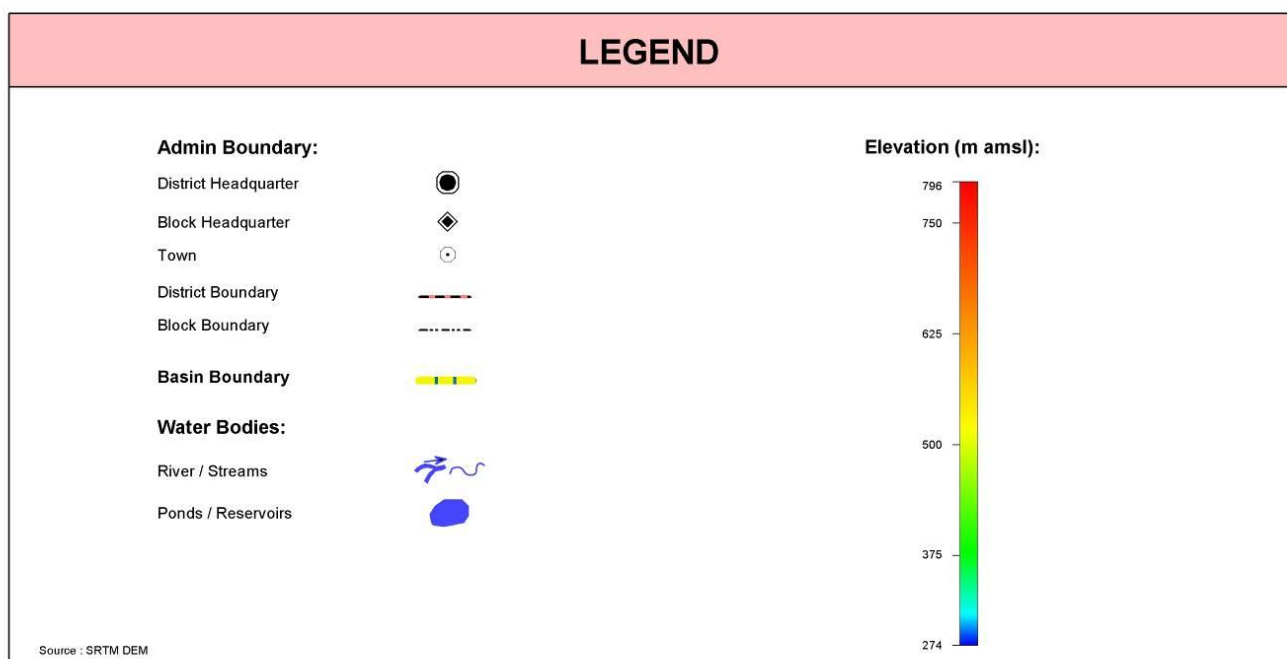
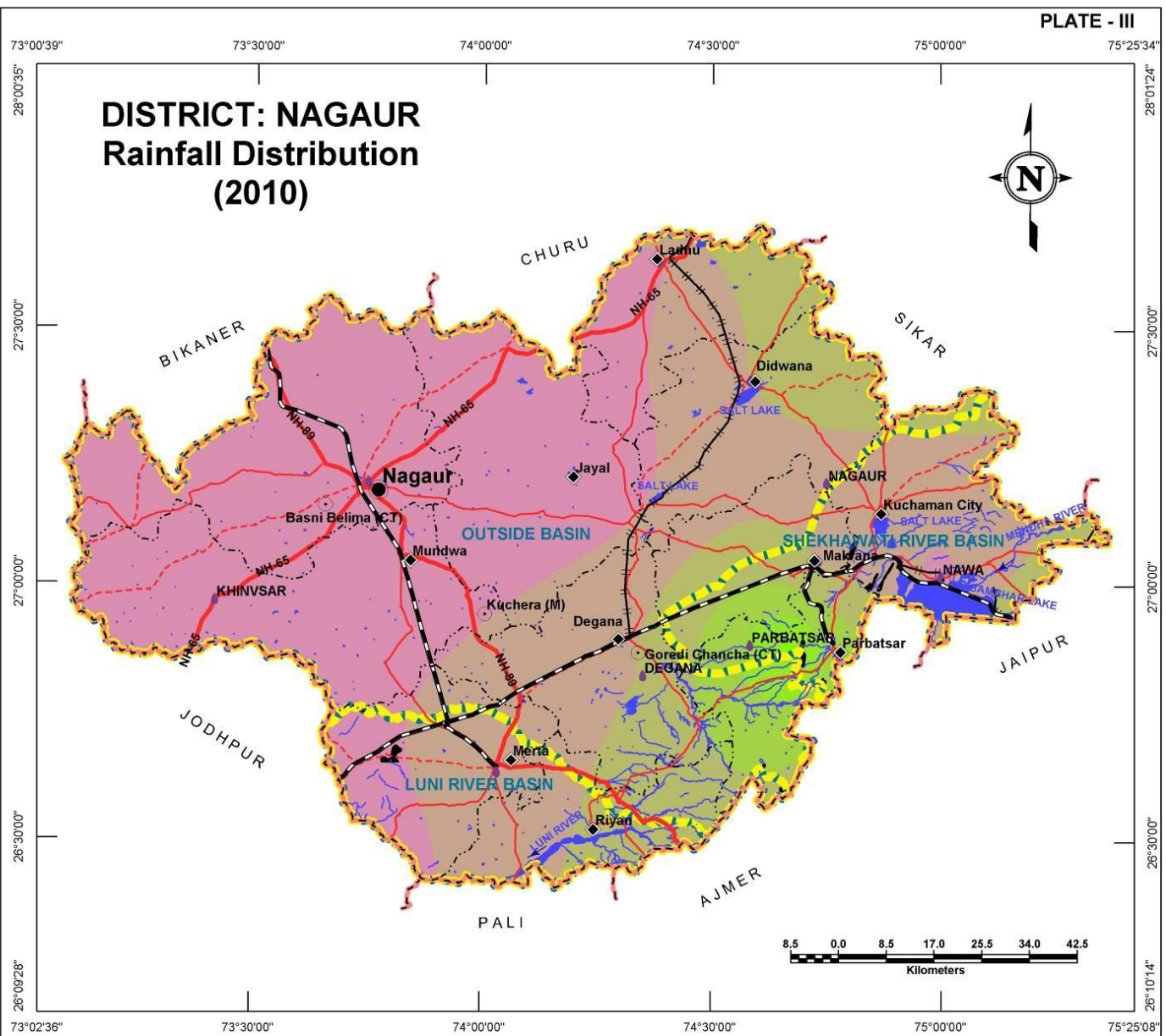
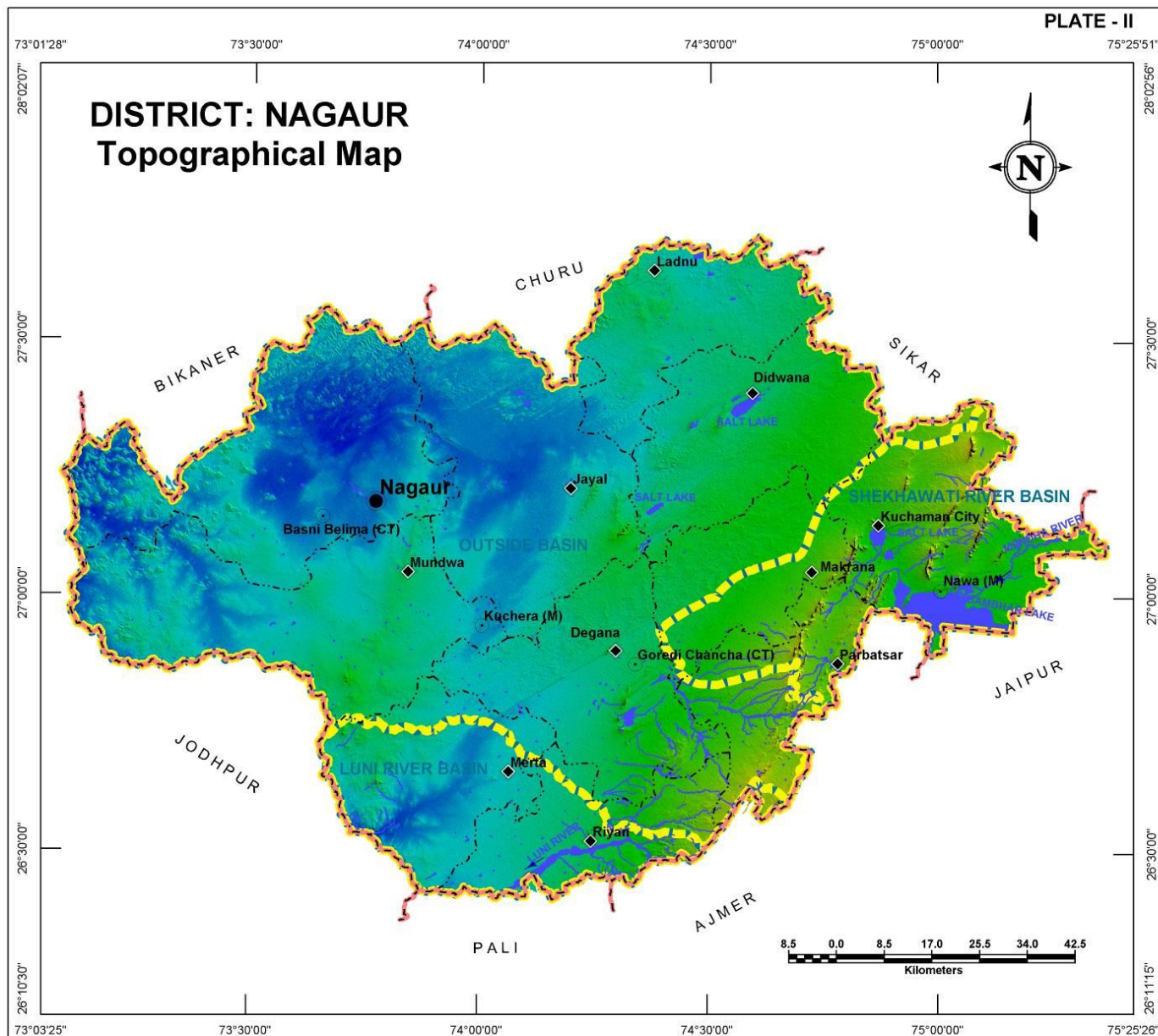
| S. No. | Block Name    | Min. Elevation<br>(m amsl) | Max. Elevation<br>(m amsl) |
|--------|---------------|----------------------------|----------------------------|
| 1      | Degana        | 299.0                      | 459.0                      |
| 2      | Didwana       | 306.0                      | 416.0                      |
| 3      | Jayal         | 284.0                      | 454.0                      |
| 4      | Kuchaman City | 357.0                      | 688.0                      |
| 5      | Ladnu         | 296.0                      | 371.0                      |
| 6      | Makrana       | 339.0                      | 633.0                      |
| 7      | Merta         | 279.0                      | 384.0                      |
| 8      | Mundwa        | 278.0                      | 391.0                      |
| 9      | Nagaur        | 275.0                      | 348.0                      |
| 10     | Parbatsar     | 365.0                      | 794.0                      |
| 11     | Riyan         | 298.0                      | 735.0                      |

## RAINFALL

The rainfall is generally scanty and erratic. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III where rainfall is gradually increasing from northwest to southeast. Most of the blocks seem to have generally received rainfall in the range of 400 to 600 mm in the year 2010 and the average annual rainfall was 559.4 mm based on the data of available blocks. Parbatsar block received highest rainfall (911.6 mm) whereas lowest was in Nagaur block (424.1 mm). Highest average annual rainfall was recorded in Parbatsar block about 750.7 mm.

**Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)**

| Block Name    | Minimum Annual<br>Rainfall (mm) | Maximum Annual<br>Rainfall (mm) | Average Annual<br>Rainfall (mm) |
|---------------|---------------------------------|---------------------------------|---------------------------------|
| Degana        | 495.7                           | 736.8                           | 579.6                           |
| Didwana       | 475.4                           | 666.7                           | 576.6                           |
| Jayal         | 434.0                           | 534.8                           | 468.5                           |
| Kuchaman City | 478.7                           | 666.7                           | 566.1                           |
| Ladnu         | 462.8                           | 680.9                           | 545.4                           |
| Makrana       | 543.3                           | 859.8                           | 635.1                           |
| Merta         | 466.5                           | 574.3                           | 511.6                           |
| Mundwa        | 436.5                           | 508.7                           | 466.1                           |
| Nagaur        | 424.1                           | 462.7                           | 445.1                           |
| Parbatsar     | 590.2                           | 911.6                           | 750.7                           |
| Riyan         | 491.4                           | 719.5                           | 608.7                           |



# GEOLOGY and GEOMORPHOLOGY

## Geology

The district is occupied by the Delhi Super Group, Erinpura granite and Marwar Super Group rocks. Delhi Super Group comprises of the Punagarh, Siroli, Kumbhalgarh, Alwar and Ajabgarh Groups mainly exposed in the Ladnu, Didwana, Makrana, Degana and Riyan blocks of the district trending NNE to south. The Kumbhalgarh Group comprises of calc-schist, gneiss and marble, exposed in Riyan and Kuchman city blocks in small patches. The Alwar Group comprises mainly quartzite and arkose whereas the overlying Ajabgarh Group is consists of crystalline limestone and marble, siliceous limestone, ferruginous quartzite, slate and schist both mainly exposed in the northeastern part of district around Kuchaman city and Parbatsar blocks. Erinpura granite & gneiss occupies the NNE part and exposed in Ladnu, Didwana, Makrana and Degana blocks. Malani Igneous Suite of rocks viz. rhyolite and granite are exposed in the NNE to SSW part of the district in Ladnu, Didwana, Degana, Merta and Riyan blocks of the district. Marwar Super Group, occupying more than 50% area of the district is mainly exposed in the western part of the district. This sedimentary sequence has been classified into Jodhpur, Bilara and Nagaur Groups and consists of sandstone, dolomite and limestone.

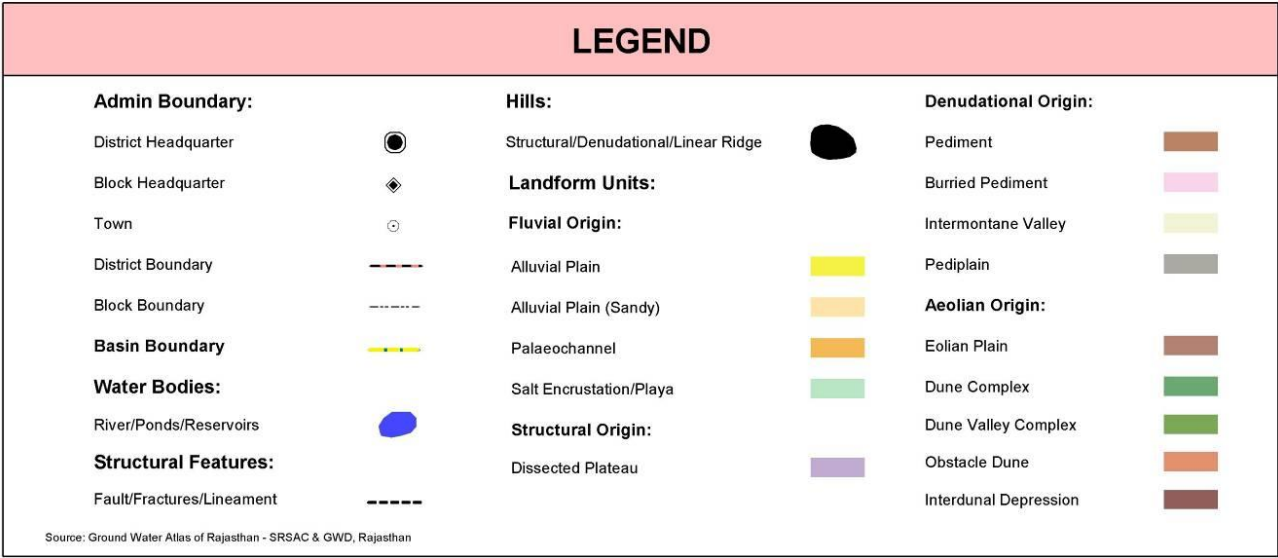
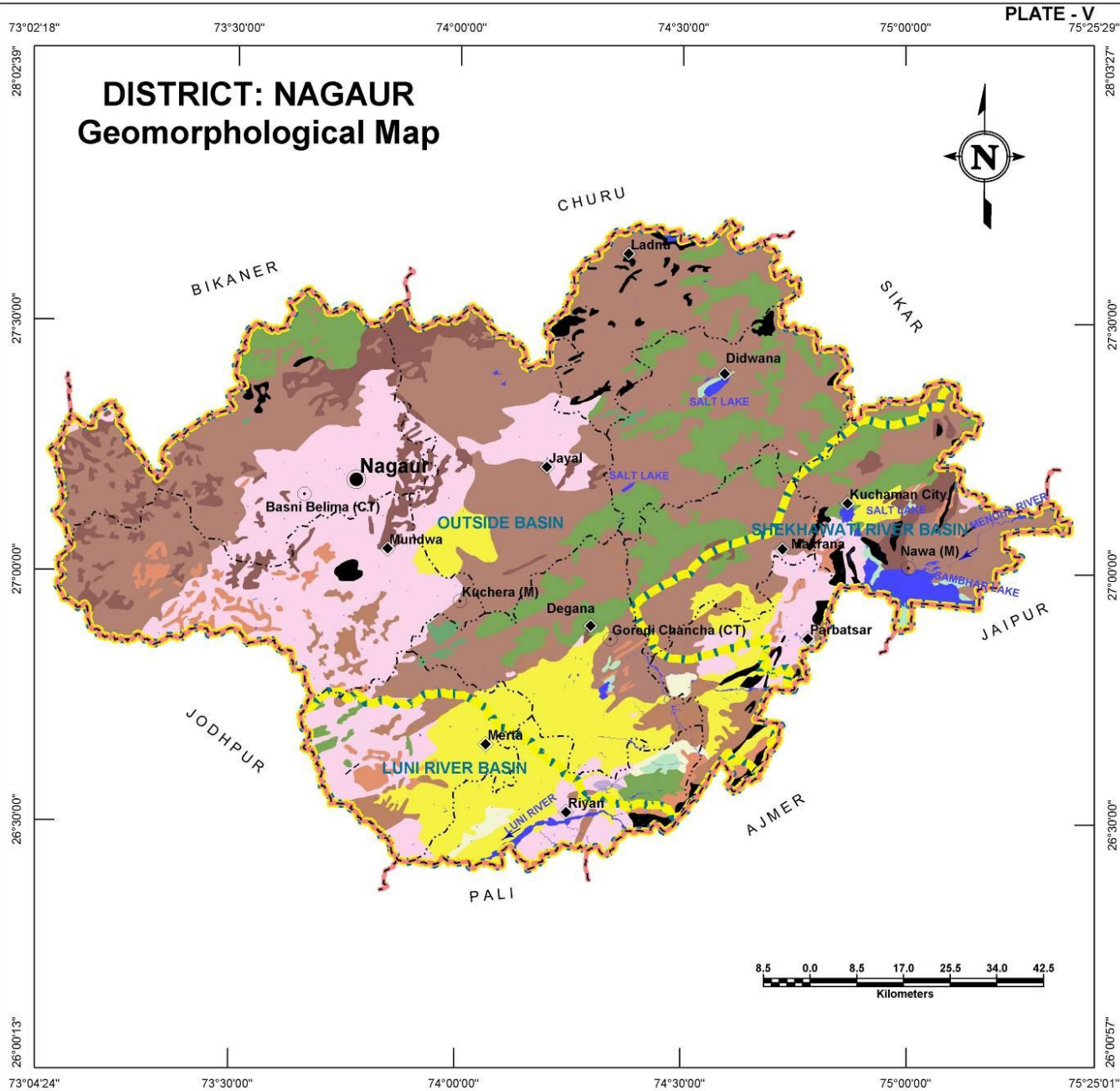
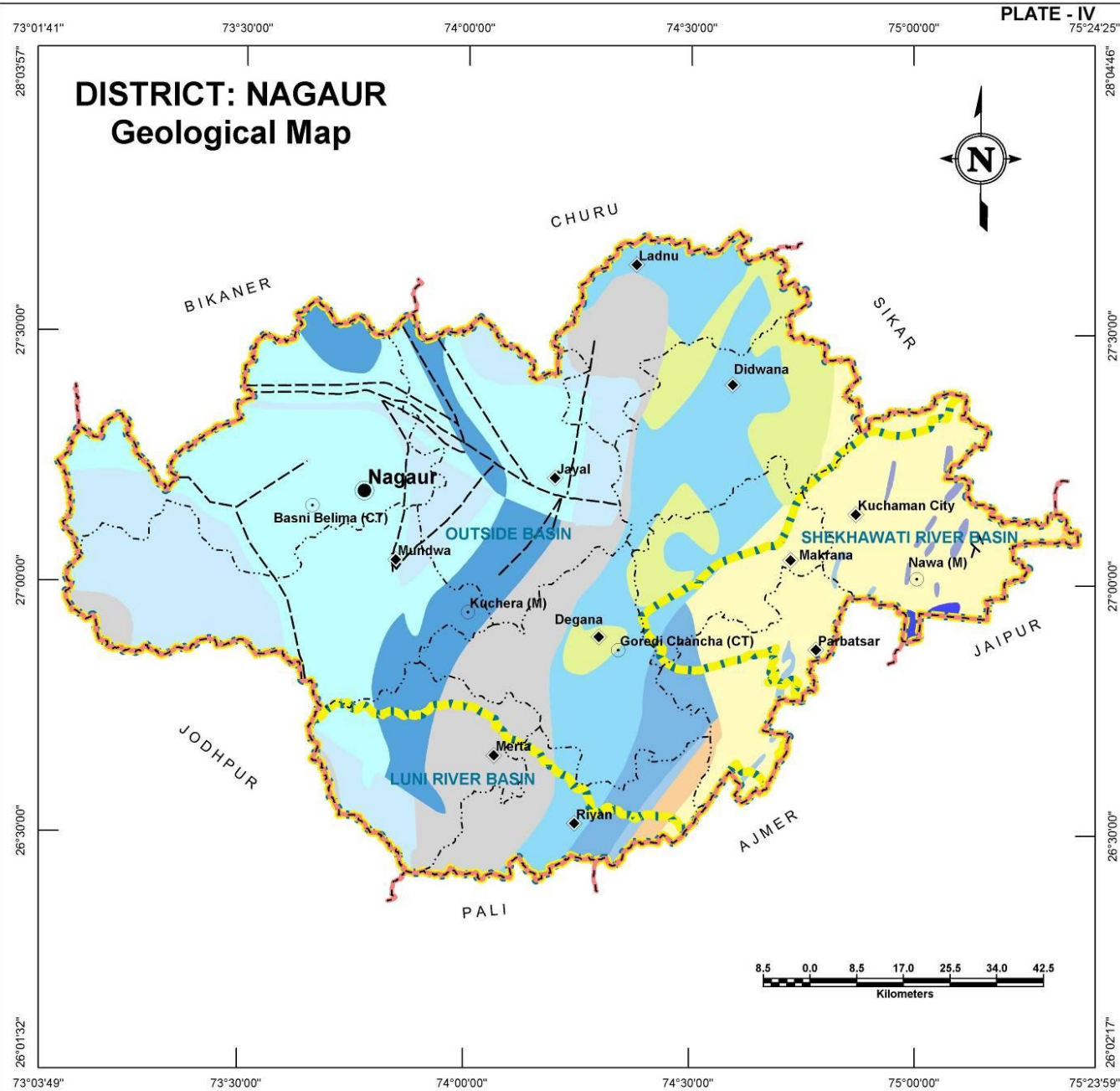
| Super Group | Group                | Formation                                                                                  |
|-------------|----------------------|--------------------------------------------------------------------------------------------|
|             | Quaternary           | Aeolian sand, Kankar, clay                                                                 |
| Marwar      | Nagaur               | Sandstone                                                                                  |
|             | Bilara               | Dolomite and Limestone                                                                     |
|             | Jodhpur              | Sandstone                                                                                  |
|             | Malani Igneous suite |                                                                                            |
| Delhi       | Punagarh             | Shale, Phyllite, micaschist & quartzite with metavolcanics                                 |
|             | Sirohi               | Phyllite, mica schist with intercalated quartzite & marble                                 |
|             | Intrusive            | Erinpura and Sendra-Ambaji Granite & Gneiss                                                |
|             | Kumbhalgarh          | Calc-schist, gneisses, marble                                                              |
|             | Ajabgarh             | Crystalline limestone & marble, siliceous limestone, ferruginous quartzite, slate & schist |
|             | Alwar                | Quartzite and arkose                                                                       |

# DISTRICT – NAGAU

## Geomorphology

**Table: Geomorphologic units, their description and distribution**

| Origin       | Landform Unit                               | Description                                                                                                                                                                                                                                                                                                                                                       |
|--------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aeolian      | Dune Complex                                | An undulating plain composed of number of sand dunes of crescent shape.                                                                                                                                                                                                                                                                                           |
|              | Dune Valley Complex                         | Cluster of dunes and interdunal spaces with undulating topography formed due to wind-blown activity, comprising of unconsolidated sand and silt.                                                                                                                                                                                                                  |
|              | Eolian Plain                                | Formed by aeolian activity, with sand dunes of varying height, size and slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation.                                                                                                                    |
|              | Interdunal Depression                       | Slightly depressed area in between the dunal complex showing moisture and fine sediments.                                                                                                                                                                                                                                                                         |
|              | Obstacle Dune                               | Formed on windward/leeward sides of obstacle like isolated hills or continuous chain of hills, dune to obstruction in path of sand laden winds. Badly dissected well cemented and vegetated.                                                                                                                                                                      |
| Denudational | Buried Pediment                             | Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.                                                                                                                                                                                                                                                                   |
|              | Intermontane Valley                         | Depression between mountains, generally broad & linear, filled with colluvial deposits.                                                                                                                                                                                                                                                                           |
|              | Pediment                                    | Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.                                                                                                                                                                                                 |
|              | Pediplain                                   | Coalescence and extensive occurrence of pediment.                                                                                                                                                                                                                                                                                                                 |
| Fluvial      | Alluvial Plain                              | Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.                                                                                                                                                                                  |
|              | Alluvial Plain (Sandy)                      | Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.                                                                                                                                                                |
|              | Paleochannel                                | Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.                                                                                                                                                                                                                                                        |
|              | Salt Encrustation/ Playa                    | Topographical depression comprising of clay, silt, sand and soluble salts, usually undrained and devoid of vegetation.                                                                                                                                                                                                                                            |
| Hills        | Denudational, Structural Hill, Linear Ridge | Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike. |



## AQUIFERS

## DISTRICT – NAGAUR

Older alluvium constitutes the most spatially prominent aquifer in the district comprising of sand, silt and kankar occupying about 37% of district area. Other aquifers are in hardrocks and important among them are limestone of Bilara Group, Tertiary sandstones and the Nagaur Jodhpur sandstone occupying about 20% and 28% aquifer areas in the district respectively. In the southeastern part of the district aquifers formed in schist is seen occupying about 2570 sq kms aquifer area i.e., 14.4%.

**Table: aquifer potential zones their area and their description**

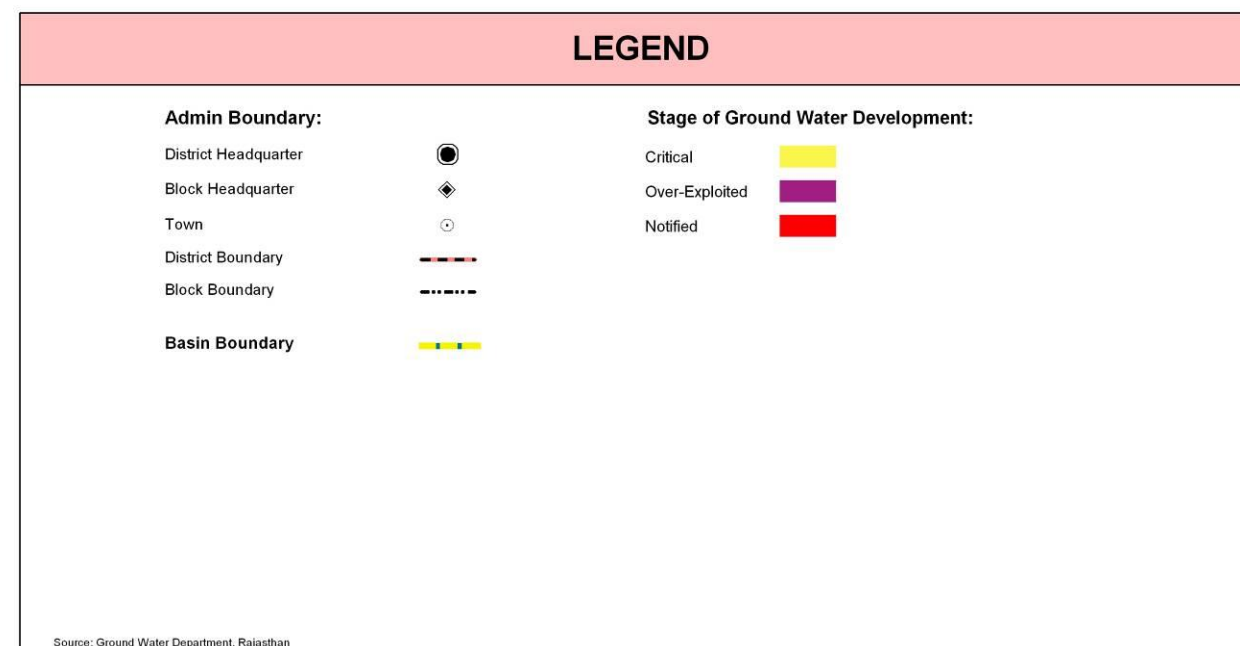
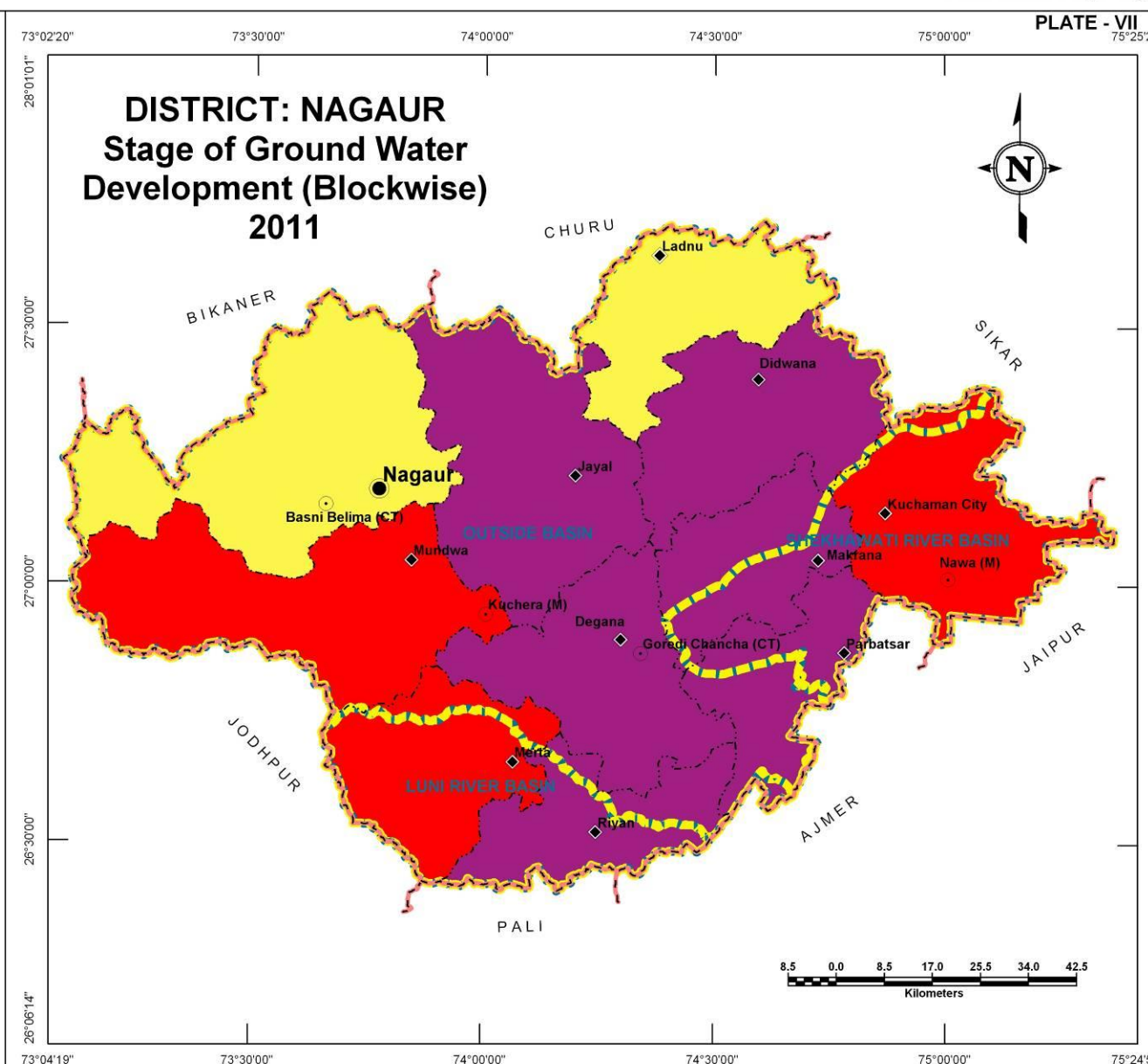
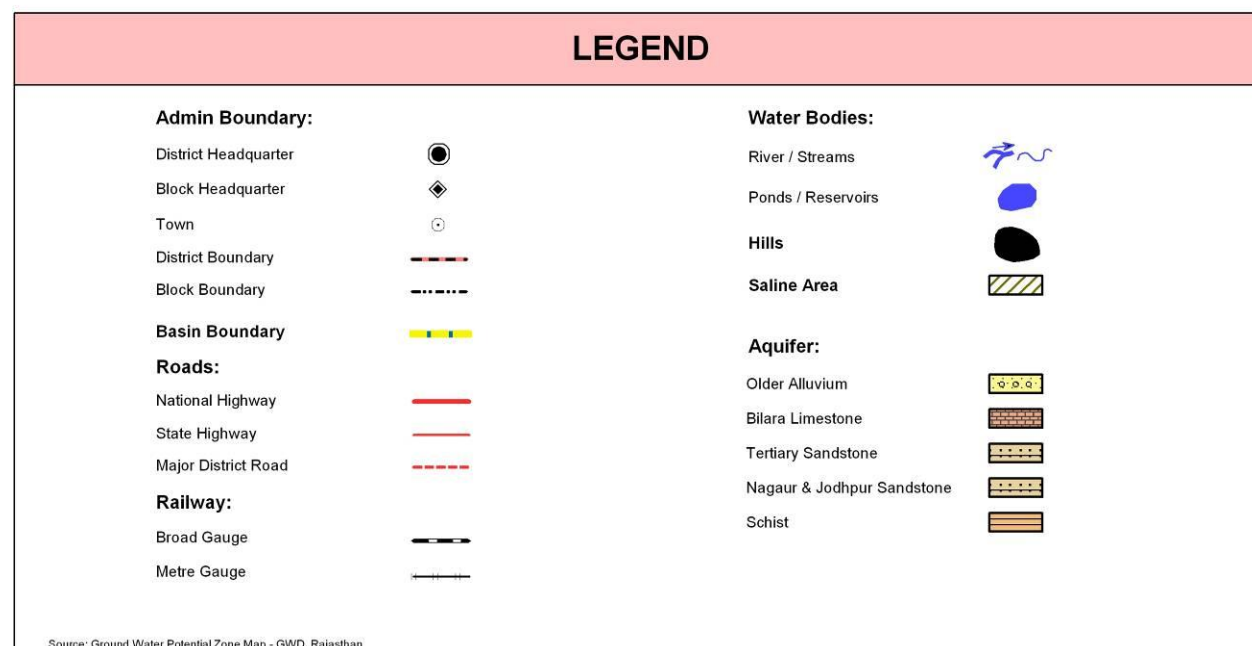
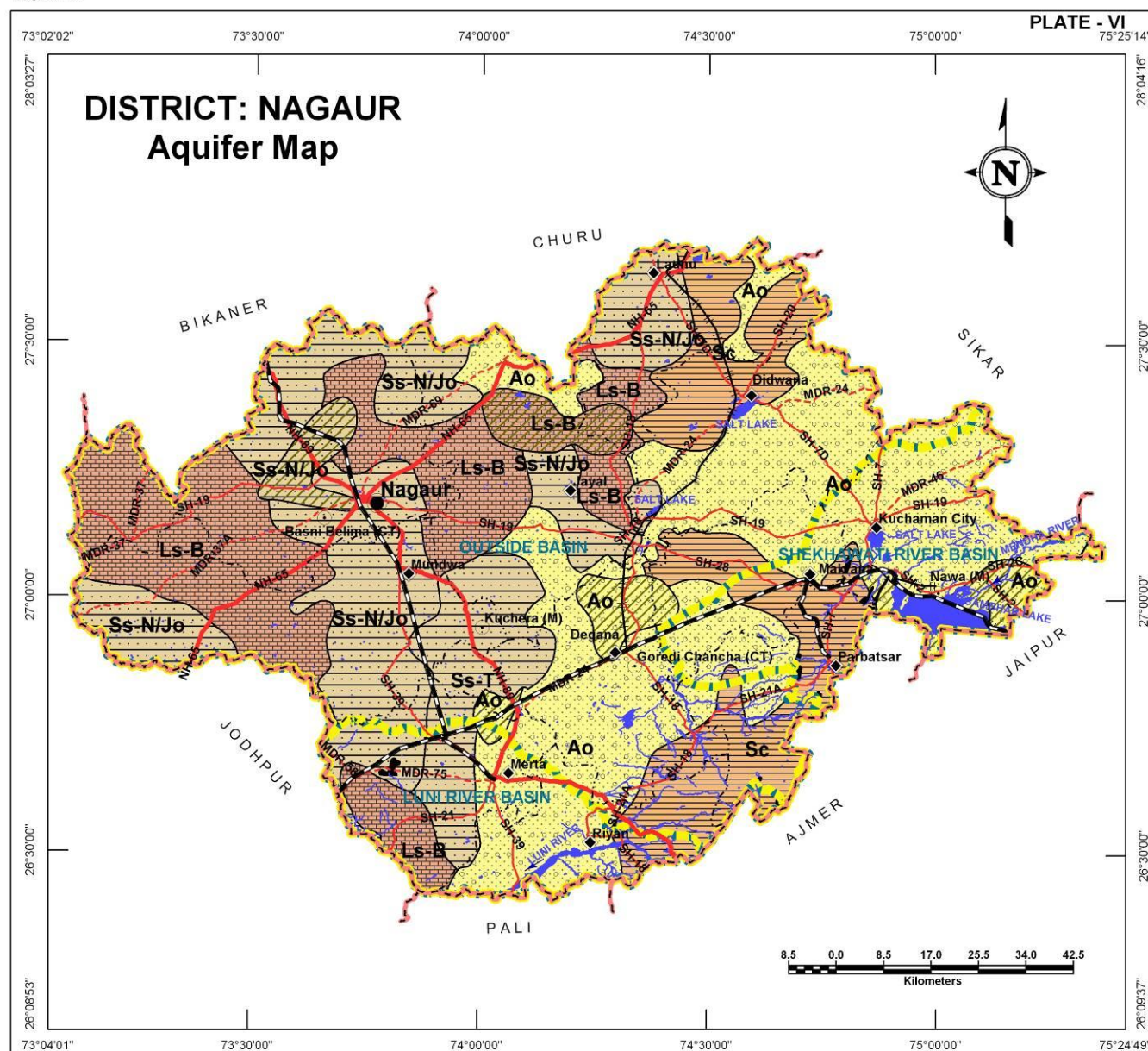
| Aquifer in Potential Zone  | Area (sq km)    | % age of district | Description of the unit/Occurrence                                                                       |
|----------------------------|-----------------|-------------------|----------------------------------------------------------------------------------------------------------|
| Older Alluvium             | 6,562.0         | 36.9              | This litho-unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.      |
| Bilara Limestone           | 3,629.8         | 20.4              | It is grey to buff coloured hard and compact.                                                            |
| Tertiary Sandstone         | 1,093.8         | 6.1               | Medium to coarse grained, consolidated to semi consolidated sandstone.                                   |
| Nagaur & Jodhpur Sandstone | 3,916.5         | 22.0              | Buff to reddish brown in colour, fine to medium grained hard and compact sandstone.                      |
| Schist                     | 2,568.5         | 14.4              | Medium to fine grained compact rock. The litho units are soft, friable and have closely spaced cleavage. |
| Hills                      | 34.5            | 0.2               |                                                                                                          |
| <b>Total</b>               | <b>17,805.1</b> | <b>100.0</b>      |                                                                                                          |

## STAGE OF GROUND WATER DEVELOPMENT

Ground water development is very high in the district as suggested by the resource assessment studies and categorization of blocks on that basis. Three blocks have been categorized as Notified where the ground water is under severe stress and no more development of it is permissible. >100% development of ground water is seen in six other blocks of the district which have been categorized as 'Over Exploited'. Ladnu and Nagaur are the two districts where the ground water situation appears to be better than other blocks but still falling within the 'Critical' category i.e., close to 100% development has already taken place.

| Categorization on the basis of stage of development of ground water | Block Name                                        |
|---------------------------------------------------------------------|---------------------------------------------------|
| Critical                                                            | Ladnu, Nagaur                                     |
| Over Exploited                                                      | Jayal, Parbatsar, Makrana, Didwana, Riyan, Degana |
| Notified                                                            | Mundwa, Merta, Kuchaman City                      |

**Basis for categorization:** Ground water development  $\leq 100\%$  - Critical and  $>100\%$  - Over-Exploited. In Notified blocks development of GW is not permitted any more.



## LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS

## DISTRICT – NAGAU

Nagaur district has a well distributed network of exploratory wells (265) and ground water monitoring stations (324) in the district owned by RGWD (184 and 311 respectively) and CGWB (81 and 13 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that ground water level monitoring network is sufficiently distributed for appropriate monitoring but for water quality, 113 wells are recommended to be added to existing network for optimum monitoring of the aquifers.

**Table: Block wise count of wells (existing and recommended)**

| Block Name    | Exploratory Wells |            |            | Ground Water Monitoring Stations |            |            | Recommended additional wells for optimization of monitoring network |               |
|---------------|-------------------|------------|------------|----------------------------------|------------|------------|---------------------------------------------------------------------|---------------|
|               | CGWB              | RGWD       | Total      | CGWB                             | RGWD       | Total      | Water Level                                                         | Water Quality |
| Degana        | 5                 | 20         | 25         | 1                                | 35         | 36         | 0                                                                   | 12            |
| Didwana       | 22                | 25         | 47         | -                                | 31         | 31         | 0                                                                   | 20            |
| Jayal         | 4                 | 19         | 23         | 2                                | 29         | 31         | 0                                                                   | 1             |
| Kuchaman City | 13                | 18         | 31         | -                                | 35         | 35         | 0                                                                   | 49            |
| Ladnu         | 11                | 13         | 24         | 1                                | 32         | 33         | 0                                                                   | 0             |
| Makrana       | 0                 | 12         | 12         | 1                                | 25         | 26         | 0                                                                   | 2             |
| Merta         | 12                | 24         | 36         | -                                | 29         | 29         | 0                                                                   | 4             |
| Mundwa        | 7                 | 20         | 27         | 3                                | 23         | 26         | 0                                                                   | 2             |
| Nagaur        | 1                 | 15         | 16         | 1                                | 35         | 36         | 0                                                                   | 0             |
| Parbatsar     | 1                 | 4          | 5          | 1                                | 18         | 19         | 0                                                                   | 9             |
| Riyan         | 5                 | 14         | 19         | 3                                | 19         | 22         | 0                                                                   | 14            |
| <b>Total</b>  | <b>81</b>         | <b>184</b> | <b>265</b> | <b>13</b>                        | <b>311</b> | <b>324</b> | <b>0</b>                                                            | <b>113</b>    |

## DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

Depth to water level shows variation from less than 10m below ground level to about 130m below bgl. The alluvial aquifers in the eastern and northern parts have shown a fairly shallower ground water occurrence i.e., upto 60m of depth from ground level whereas the hardrock areas in the western part i.e., within Sandstone and limestone aquifers very deep water levels reaching upto 130m bgl are seen.

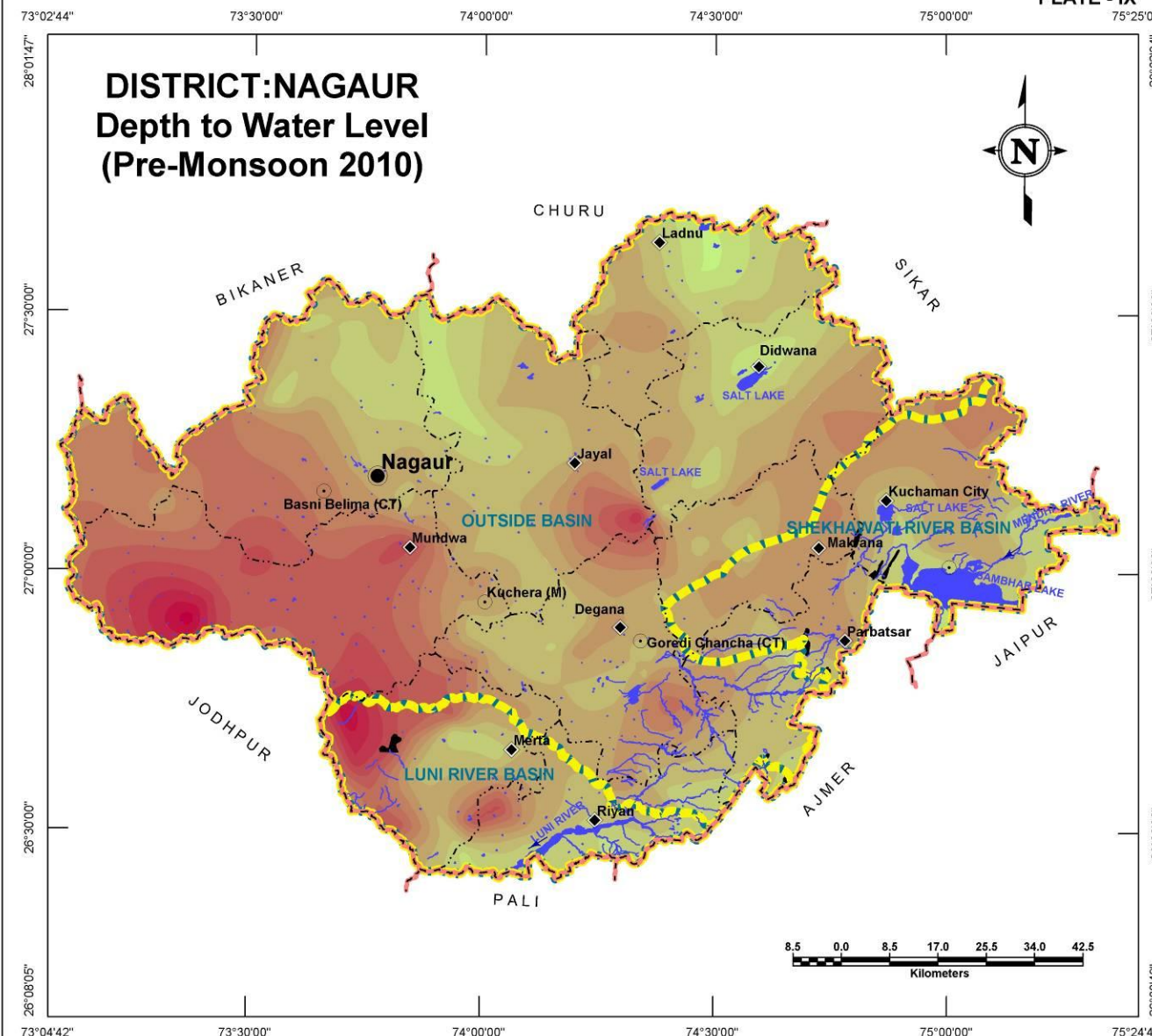
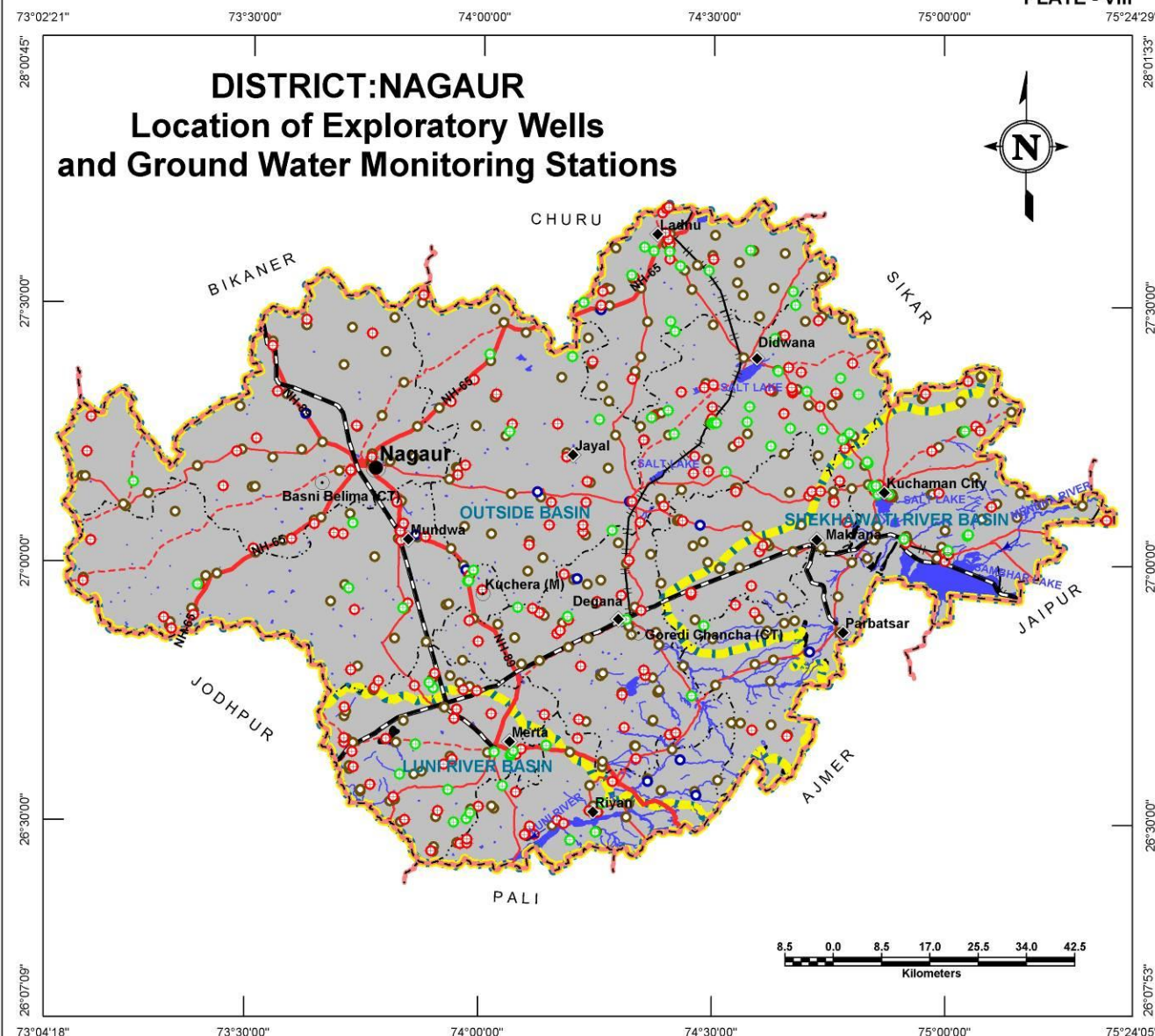
| Block Name    | Block wise area coverage (sq km) per depth to water level range (m bgl) * |              |                |                |                |                |                |                |                |              |              |              |             |            | Total Area (sq km) |
|---------------|---------------------------------------------------------------------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------|--------------|--------------|-------------|------------|--------------------|
|               | <10                                                                       | 10-20        | 20-30          | 30-40          | 40-50          | 50-60          | 60-70          | 70-80          | 80-90          | 90-100       | 100-110      | 110-120      | 120-130     | >130       |                    |
| Degana        | -                                                                         | -            | 3.0            | 387.3          | 597.9          | 363.5          | 159.3          | 59.3           | 8.3            | -            | -            | -            | -           | -          | 1,578.6            |
| Didwana       | 3.0                                                                       | 77.3         | 291.6          | 446.4          | 441.9          | 280.3          | 97.2           | 3.6            | 0.5            | -            | -            | -            | -           | -          | 1,641.8            |
| Jayal         | -                                                                         | 162.8        | 284.7          | 677.9          | 671.3          | 181.7          | 87.0           | 67.2           | 42.4           | 21.5         | 4.7          | -            | -           | -          | 2,201.2            |
| Kuchaman City | -                                                                         | 56.8         | 364.9          | 280.7          | 332.5          | 399.0          | 191.2          | 0.3            | -              | -            | -            | -            | -           | -          | 1,625.4            |
| Ladnu         | 32.9                                                                      | 161.5        | 272.8          | 327.4          | 386.3          | 94.1           | 0.3            | -              | -              | -            | -            | -            | -           | -          | 1,275.3            |
| Makrana       | -                                                                         | -            | 6.2            | 81.0           | 394.2          | 551.5          | 40.5           | 1.0            | -              | -            | -            | -            | -           | -          | 1,074.4            |
| Merta         | -                                                                         | -            | 54.9           | 177.8          | 287.5          | 170.3          | 116.8          | 163.1          | 171.6          | 78.9         | 62.6         | 48.9         | 17.1        | 0.1        | 1,349.6            |
| Mundwa        | -                                                                         | -            | -              | 25.5           | 78.5           | 122.6          | 212.9          | 418.7          | 930.1          | 305.3        | 123.8        | 75.7         | 36.4        | 6.0        | 2,335.5            |
| Nagaur        | -                                                                         | 3.1          | 337.0          | 529.3          | 244.7          | 223.5          | 623.7          | 470.4          | 107.6          | 15.6         | -            | -            | -           | -          | 2,554.9            |
| Parbatsar     | -                                                                         | -            | 83.5           | 392.5          | 275.2          | 273.9          | 2.1            | -              | -              | -            | -            | -            | -           | -          | 1,027.2            |
| Riyan         | -                                                                         | 34.4         | 346.8          | 308.6          | 289.1          | 54.3           | 28.9           | 23.4           | 18.5           | 2.7          | -            | -            | -           | -          | 1,106.7            |
| <b>Total</b>  | <b>35.9</b>                                                               | <b>495.9</b> | <b>2,045.4</b> | <b>3,634.4</b> | <b>3,999.1</b> | <b>2,714.7</b> | <b>1,559.9</b> | <b>1,207.0</b> | <b>1,279.0</b> | <b>424.0</b> | <b>191.1</b> | <b>124.6</b> | <b>53.5</b> | <b>6.1</b> | <b>17,770.6</b>    |

\* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.



PLATE - VIII

PLATE - IX



**LEGEND**

**Admin Boundary:**

District Headquarter

Block Headquarter

Town

District Boundary

Block Boundary

Basin Boundary

**Roads:**

National Highway

State Highway

Major District Road

**Railway:**

Broad Gauge

Metre Gauge

**Water Bodies:**

River / Streams

Ponds / Reservoirs

Hills

**Exploratory Wells:**

CGWB

RGWD

**Ground Water Monitoring Stations:**

CGWB

RGWD

**LEGEND**

**Admin Boundary:**

District Headquarter

Block Headquarter

Town

District Boundary

Block Boundary

Basin Boundary

**Water Bodies:**

River / Streams

Ponds / Reservoirs

Hills

**Depth to Water Level (m bgl):**

<10

10 - 20

20 - 30

30 - 40

40 - 50

50 - 60

60 - 70

70 - 80

80 - 90

90 - 100

100 - 110

110 - 120

120 - 130

> 130

## WATER TABLE ELEVATION (PRE MONSOON – 2010)

## DISTRICT – NAGAU

The flow direction of ground water in the district is broadly indicated as from east to west with local modification occasionally. Perusal of Plate – X indicates that water table elevation is highest reaching up to >440m above mean sea level in southeast part (Parbatsar Block) of the district whereas minimum elevation (<180m amsl) is in the Mundwa and Merta blocks in the western most part of the district.

Table: Block wise area covered in water table elevation range

| Block Name    | Block wise area coverage (sq km) per water table elevation range (m amsl) |              |              |                |                |                |                |                |                |              |                |              |              |             | Total Area (sq km) |
|---------------|---------------------------------------------------------------------------|--------------|--------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------|----------------|--------------|--------------|-------------|--------------------|
|               | < 180                                                                     | 180 - 200    | 200 - 220    | 220 - 240      | 240 - 260      | 260 - 280      | 280 - 300      | 300 - 320      | 320 - 340      | 340 - 360    | 360 - 380      | 380 - 400    | 400 - 440    | > 440       |                    |
| Degana        | -                                                                         | -            | -            | -              | 74.4           | 214.3          | 339.3          | 555.3          | 198.2          | 103.7        | 66.8           | 26.6         | -            | -           | 1,578.6            |
| Didwana       | -                                                                         | -            | -            | -              | 2.9            | 126.6          | 336.4          | 548.6          | 495.9          | 130.0        | 1.4            | -            | -            | -           | 1,641.8            |
| Jayal         | -                                                                         | -            | -            | -              | 1,553.2        | 579.3          | 68.2           | 0.5            | -              | -            | -              | -            | -            | -           | 2,201.2            |
| Kuchaman City | -                                                                         | -            | -            | -              | -              | -              | -              | -              | 25.2           | 333.3        | 952.6          | 235.0        | 79.3         | -           | 1,625.4            |
| Ladnu         | -                                                                         | -            | -            | -              | 106.3          | 260.0          | 390.3          | 513.7          | 5.0            | -            | -              | -            | -            | -           | 1,275.3            |
| Makrana       | -                                                                         | -            | -            | -              | -              | -              | -              | 403.9          | 450.2          | 153.6        | 49.8           | 16.9         | -            | -           | 1,074.4            |
| Merta         | -                                                                         | -            | 67.1         | 598.4          | 295.7          | 197.4          | 191.0          | -              | -              | -            | -              | -            | -            | -           | 1,349.6            |
| Mundwa        | 1.8                                                                       | 205.9        | 513.8        | 822.3          | 713.5          | 78.2           | -              | -              | -              | -            | -              | -            | -            | -           | 2,335.5            |
| Nagaur        | -                                                                         | -            | 416.0        | 1,026.6        | 761.3          | 351.0          | -              | -              | -              | -            | -              | -            | -            | -           | 2,554.9            |
| Parbatsar     | -                                                                         | -            | -            | -              | -              | -              | -              | 30.5           | 71.7           | 81.5         | 188.5          | 353.4        | 282.0        | 19.6        | 1,027.2            |
| Riyan         | -                                                                         | -            | -            | 56.0           | 99.9           | 72.5           | 143.2          | 229.8          | 132.7          | 88.0         | 78.7           | 89.9         | 115.0        | 1.0         | 1,106.7            |
| <b>Total</b>  | <b>1.8</b>                                                                | <b>205.9</b> | <b>996.9</b> | <b>2,503.3</b> | <b>3,607.2</b> | <b>1,879.3</b> | <b>1,468.4</b> | <b>2,282.3</b> | <b>1,378.9</b> | <b>890.1</b> | <b>1,337.8</b> | <b>721.8</b> | <b>476.3</b> | <b>20.6</b> | <b>17,770.6</b>    |

## WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

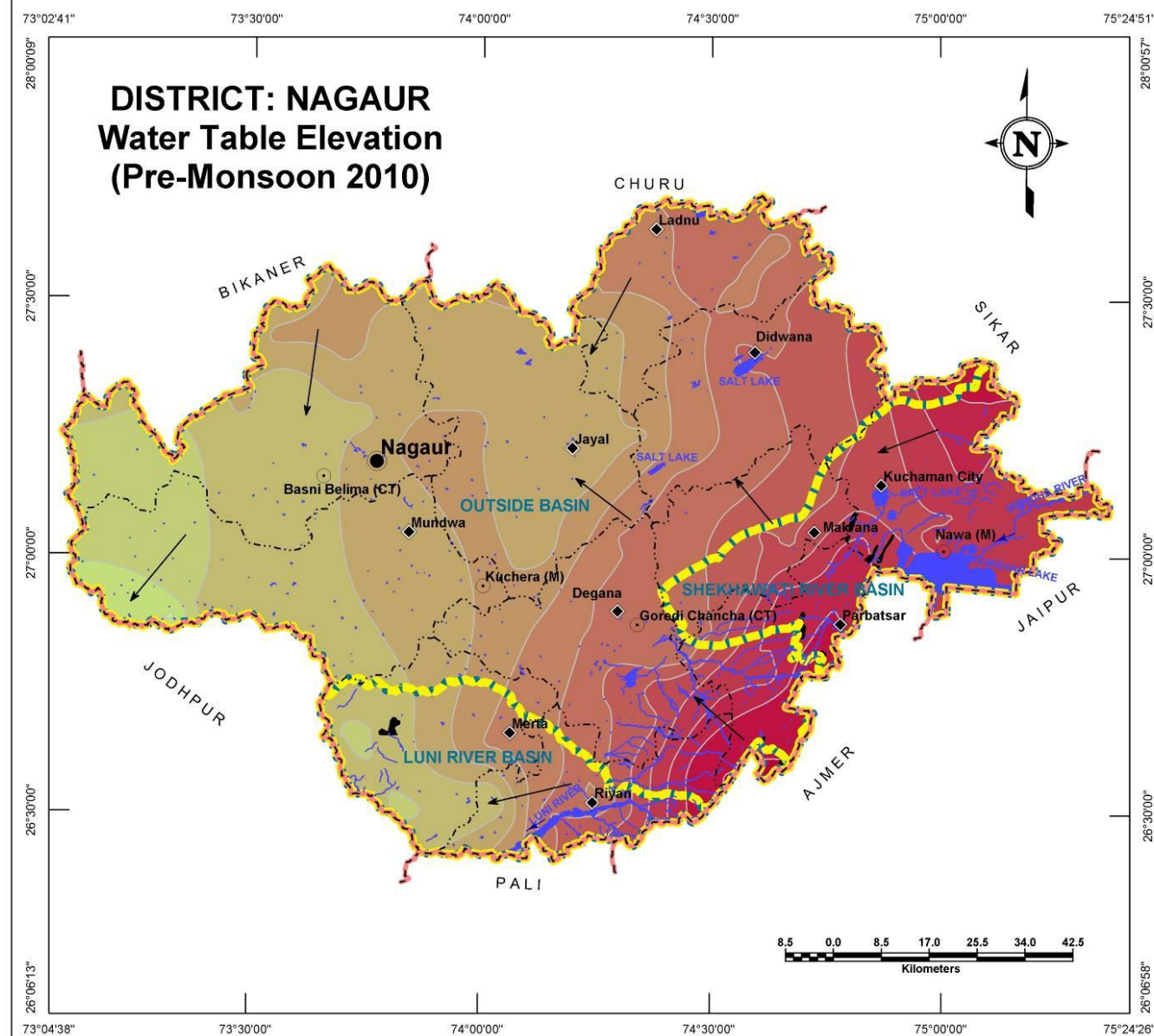
Ground water level fluctuation is seen to vary from fall of 2m to rise by 16m in the district as seen in Plate – XI. The negative fluctuation areas, mostly in the range from 0 to -2m are the areas where over exploitation is taking place as a belt running from west to east in the central part of the district. Rest of the district has shown a general rise in ground water level in the post monsoon season by upto 4m and occasionally rising by 16m (however very localized rise in nature) as seen in southwestern part of Didwana block.

Table: Block wise area covered in water fluctuation zone

| Water level fluctuation range (m) | Block wise area coverage (sq km) |                |                |                |                |                |                |                |                |                |                | Total Area (sq km) |
|-----------------------------------|----------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------------|
|                                   | Degana                           | Didwana        | Jayal          | Kuchaman City  | Ladnu          | Makrana        | Merta          | Mundwa         | Nagaur         | Parbatsar      | Riyan          |                    |
| < -2                              | -                                | -              | -              | -              | -              | -              | -              | -              | 1.0            | -              | -              | 1.0                |
| -2 to 0                           | 387.5                            | 186.9          | 300.4          | 316.1          | 83.9           | 451.6          | 576.2          | 1,956.9        | 594.3          | 139.8          | 127.0          | 5,120.6            |
| 0 to 2                            | 1,191.1                          | 1,212.8        | 1,710.7        | 1,276.5        | 1,156.4        | 622.8          | 773.0          | 378.6          | 1,927.6        | 628.5          | 842.9          | 11,720.9           |
| 2 to 4                            | -                                | 169.6          | 138.2          | 30.0           | 35.0           | -              | 0.4            | -              | 32.0           | 137.0          | 88.9           | 631.1              |
| 4 to 6                            | -                                | 24.0           | 24.0           | 2.8            | -              | -              | -              | -              | -              | 75.6           | 37.8           | 164.2              |
| 6 to 8                            | -                                | 16.7           | 13.7           | -              | -              | -              | -              | -              | -              | 40.5           | 8.3            | 79.2               |
| 8 to 10                           | -                                | 12.9           | 7.4            | -              | -              | -              | -              | -              | -              | 5.8            | 1.8            | 27.9               |
| 10 to 12                          | -                                | 8.8            | 4.5            | -              | -              | -              | -              | -              | -              | -              | -              | 13.3               |
| 12 to 14                          | -                                | 5.7            | 2.0            | -              | -              | -              | -              | -              | -              | -              | -              | 7.7                |
| 14 to 16                          | -                                | 4.0            | 0.3            | -              | -              | -              | -              | -              | -              | -              | -              | 4.3                |
| > 16                              | -                                | 0.4            | -              | -              | -              | -              | -              | -              | -              | -              | -              | 0.4                |
| <b>Total</b>                      | <b>1,578.6</b>                   | <b>1,641.8</b> | <b>2,201.2</b> | <b>1,625.4</b> | <b>1,275.3</b> | <b>1,074.4</b> | <b>1,349.6</b> | <b>2,335.5</b> | <b>2,554.9</b> | <b>1,027.2</b> | <b>1,106.7</b> | <b>17,770.6</b>    |



PLATE - X



**LEGEND**

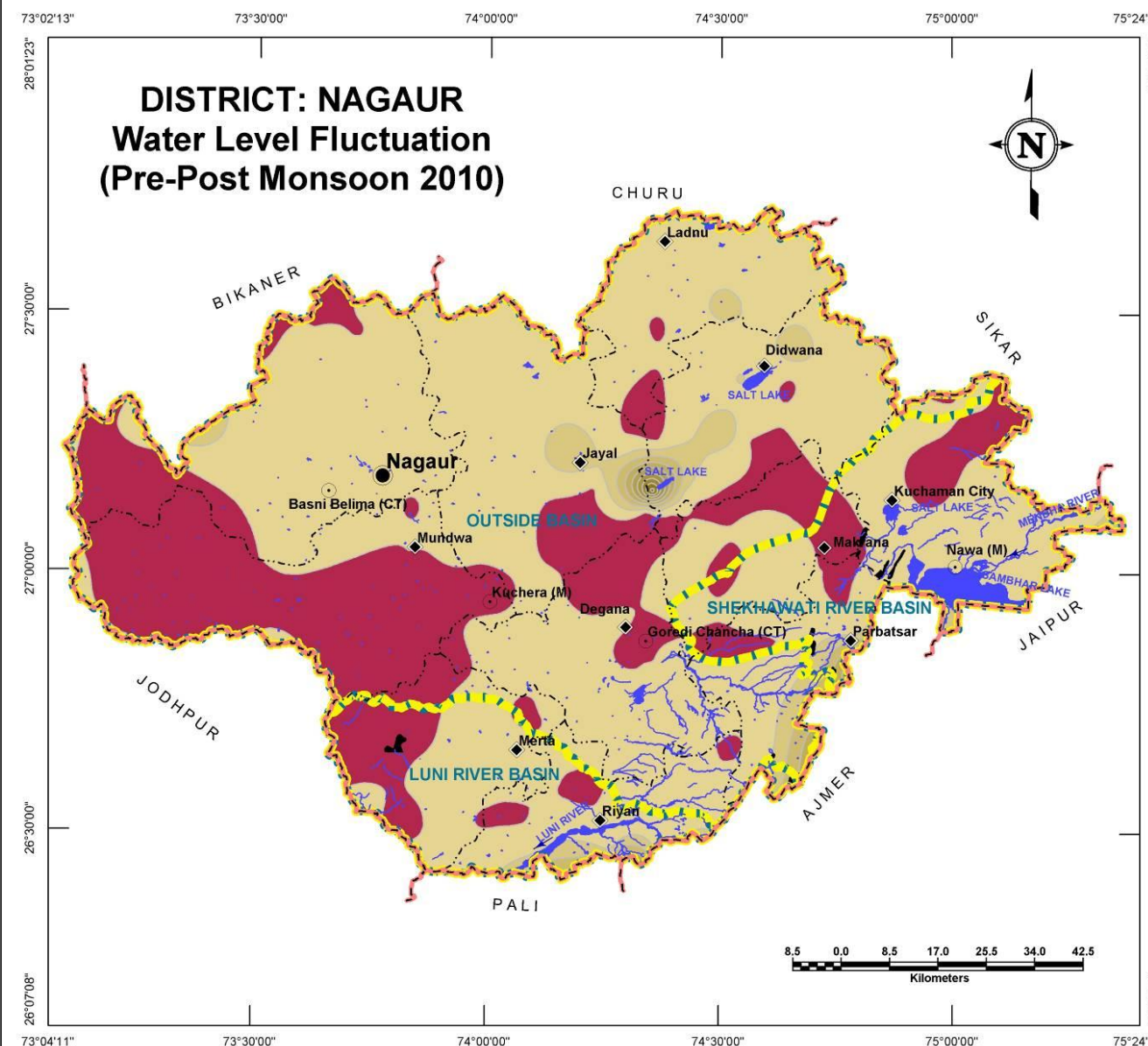
**Admin Boundary:**

- District Headquarter
- Block Headquarter
- Town
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
  - River / Streams
  - Ponds / Reservoirs
  - Hills
- Water Table Elevation

**Water Table Elevation (m amsl):**

|           |           |
|-----------|-----------|
| < 180     | 300 - 320 |
| 180 - 200 | 320 - 340 |
| 200 - 220 | 340 - 360 |
| 220 - 240 | 360 - 380 |
| 240 - 260 | 380 - 400 |
| 260 - 280 | 400 - 440 |
| 280 - 300 | >440      |

PLATE - XI



**LEGEND**

**Admin Boundary:**

- District Headquarter
- Block Headquarter
- Town
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
  - River / Streams
  - Ponds / Reservoirs
  - Hills

**Water Level Fluctuation (m):**

|          |
|----------|
| < -2     |
| -2 to 0  |
| 0 to 2   |
| 2 to 4   |
| 4 to 6   |
| 6 to 8   |
| 8 to 10  |
| 10 to 12 |
| 12 to 14 |
| 14 to 16 |
| > 16     |

## GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

## DISTRICT – NAGAU

The Electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with high EC values in ground water (>4000 µS/cm) are shown in red color and are seen to occupy almost 50% of the district in central part of the area where the ground water is not suitable for domestic purpose. Fringing the high EC areas are the areas with moderately high EC values (2000-4000 µS/cm) which also occupy significantly large areas to the extent of 38% of the district. Remaining part of the district, approximately 11% has low EC values in ground water (<2000 µS/cm) as shown in yellow color in the northern part of Kuchman City and in the western part of Parbatsar blocks. The ground water in this low EC area is suitable for domestic purpose but is of limited spatial occurrence. By and large the ground water in the district is having high EC with limited suitability for domestic purposes.

**Table: Block wise area of Electrical conductivity distribution**

| Electrical Conductivity Ranges<br>(µS/cm at 25°C)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |         |       |               |       |         |       |         |       |         |       |         |       |         |       |           |       |         |       | Total Area<br>(sq km) |
|------------------------------------------------------------------------------|----------------------------------|-------|---------|-------|---------|-------|---------------|-------|---------|-------|---------|-------|---------|-------|---------|-------|---------|-------|-----------|-------|---------|-------|-----------------------|
|                                                                              | Degana                           |       | Didwana |       | Jayal   |       | Kuchaman City |       | Ladnu   |       | Makrana |       | Merta   |       | Mundwa  |       | Nagaur  |       | Parbatsar |       | Riyan   |       |                       |
|                                                                              | Area                             | %age  | Area    | %age  | Area    | %age  | Area          | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area      | %age  | Area    | %age  |                       |
| < 2000                                                                       | 3.8                              | 0.2   | 96.0    | 5.9   | -       | -     | 550.7         | 33.9  | 44.6    | 3.5   | 56.4    | 5.3   | 227.0   | 16.8  | 501.9   | 21.5  | 5.0     | 0.2   | 329.4     | 32.1  | 176.4   | 16.0  | 1,991.2               |
| 2000-4000                                                                    | 487.9                            | 30.9  | 643.2   | 39.1  | 324.4   | 14.7  | 443.8         | 27.3  | 558.1   | 43.7  | 528.3   | 49.1  | 454.1   | 33.7  | 1,466.8 | 62.8  | 595.8   | 23.3  | 608.8     | 59.2  | 699.0   | 63.0  | 6,810.2               |
| >4000                                                                        | 1,086.9                          | 68.9  | 902.6   | 55.0  | 1,876.8 | 85.3  | 630.9         | 38.8  | 672.6   | 52.8  | 489.7   | 45.6  | 668.5   | 49.5  | 366.8   | 15.7  | 1,954.1 | 76.5  | 89.0      | 8.7   | 231.3   | 21.0  | 8,969.2               |
| Total                                                                        | 1,578.6                          | 100.0 | 1,641.8 | 100.0 | 2,201.2 | 100.0 | 1,625.4       | 100.0 | 1,275.3 | 100.0 | 1,074.4 | 100.0 | 1,349.6 | 100.0 | 2,335.5 | 100.0 | 2,554.9 | 100.0 | 1,027.2   | 100.0 | 1,106.7 | 100.0 | 17,770.6              |

## GROUND WATER CHLORIDE DISTRIBUTION

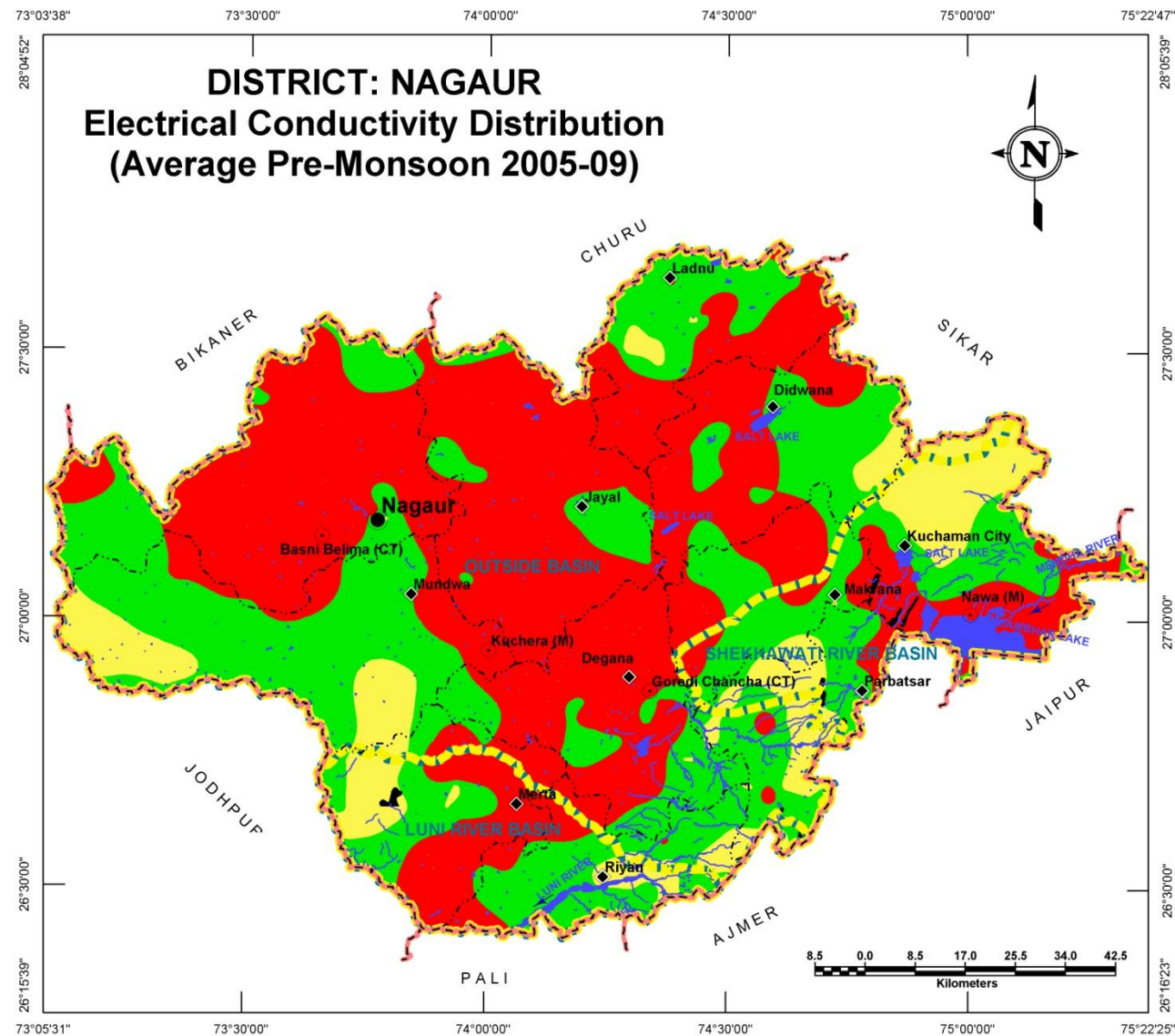
High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. The green colored regions in Plate – XIII are such areas where chloride concentration is moderately high (250-1000 mg/l) which occupy approximately 56% of the district area. The areas with high chloride concentration (>1000 mg/l) are shown in red color approximately occupying 37% of the district area. Together these two areas constitute about 93% area of the district where the ground water has limited suitability for domestic purposes. That leaves just about 7% of the district which has low chloride concentration (<250 mg/l, yellow color in the map) located around north of Kuchman City that is suitable for domestic purposes.

**Table: Block wise area of Chloride distribution**

| Chloride Concentration<br>Range (mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |         |       |               |       |         |       |         |       |         |       |         |       |         |       |           |       |         |       | Total<br>Area<br>(sq km) |
|-------------------------------------------------------------------|----------------------------------|-------|---------|-------|---------|-------|---------------|-------|---------|-------|---------|-------|---------|-------|---------|-------|---------|-------|-----------|-------|---------|-------|--------------------------|
|                                                                   | Degana                           |       | Didwana |       | Jayal   |       | Kuchaman City |       | Ladnu   |       | Makrana |       | Merta   |       | Mundwa  |       | Nagaur  |       | Parbatsar |       | Riyan   |       |                          |
|                                                                   | Area                             | %age  | Area    | %age  | Area    | %age  | Area          | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area      | %age  | Area    | %age  |                          |
| < 250                                                             | 2.4                              | -     | 157.0   | 10.0  | -       | -     | 561.8         | 34.5  | 17.4    | 1.0   | 61.9    | 6.0   | 119.8   | 9.0   | 187.2   | 8.0   | -       | -     | 167.0     | 16.5  | 82.7    | 7.5   | 1,357.2                  |
| 250-1000                                                          | 704.6                            | 45.0  | 1,092.0 | 67.5  | 451.7   | 21.0  | 484.7         | 30.0  | 798.2   | 63.0  | 742.8   | 69.0  | 812.0   | 60.0  | 1,921.7 | 82.0  | 1,135.7 | 44.0  | 794.1     | 77.5  | 931.1   | 84.2  | 9,868.6                  |
| > 1000                                                            | 871.6                            | 55.0  | 392.8   | 22.5  | 1,749.5 | 79.0  | 578.9         | 35.5  | 459.7   | 36.0  | 269.7   | 25.0  | 417.8   | 31.0  | 226.6   | 10.0  | 1,419.2 | 56.0  | 66.1      | 6.0   | 92.9    | 8.3   | 6,544.8                  |
| Total                                                             | 1,578.6                          | 100.0 | 1,641.8 | 100.0 | 2,201.2 | 100.0 | 1,625.4       | 100.0 | 1,275.3 | 100.0 | 1,074.4 | 100.0 | 1,349.6 | 100.0 | 2,335.5 | 100.0 | 2,554.9 | 100.0 | 1,027.2   | 100.0 | 1,106.7 | 100.0 | 17,770.6                 |



PLATE - XII



**LEGEND**

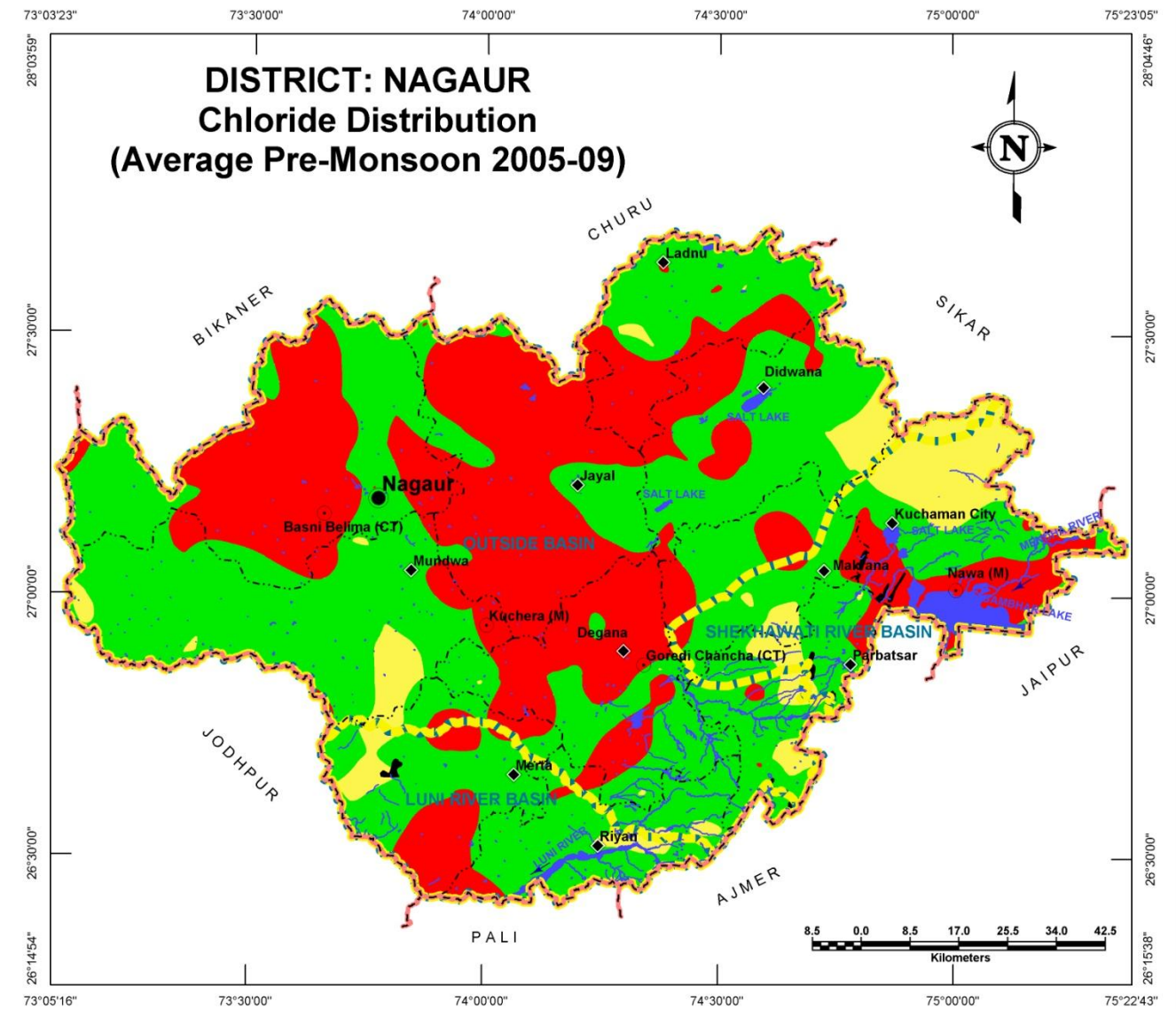
**Admin Boundary:**

- District Headquarter
- Block Headquarter
- Town
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

**Electrical Conductivity ( $\mu\text{S}/\text{cm}$  at  $25^\circ\text{C}$ ):**

- < 2000
- 2000-4000
- > 4000

PLATE - XIII



**LEGEND**

**Admin Boundary:**

- District Headquarter
- Block Headquarter
- Town
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

**Chloride Concentration (mg/l):**

- < 250
- 250 - 1000
- > 1000

## GROUND WATER FLUORIDE DISTRIBUTION

## DISTRICT – NAGOUR

The Fluoride concentration map is presented in Plate – XIV. The areas with moderately high concentration (1.5-3.0 mg/l) are shown in green color that occupy approximately 54% of the total district area and the areas with high concentration (>3.0 mg/l) are shown in red color which occupy 25% of the district area. Together these areas account for about 79% of district area where ground water is unsuitable for domestic purposes because of high (>1.5 mg/l) fluoride concentration. Remaining part (21%) of the district area falls under low concentration (<1.5 mg/l) and shown in the map with yellow color. Such areas are seen largely in the southern part of Ladnu and around the Kuchman City and Nawa where the ground water is suitable for domestic purpose.

**Table: Block wise area of Fluoride distribution**

| Fluoride concentration<br>Range(mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |         |       |               |       |         |       |         |       |         |       |         |       |         |       |           |       |         |       | Total Area<br>(sq km) |
|------------------------------------------------------------------|----------------------------------|-------|---------|-------|---------|-------|---------------|-------|---------|-------|---------|-------|---------|-------|---------|-------|---------|-------|-----------|-------|---------|-------|-----------------------|
|                                                                  | Degana                           |       | Didwana |       | Jayal   |       | Kuchaman City |       | Ladnu   |       | Makrana |       | Merta   |       | Mundwa  |       | Nagaur  |       | Parbatsar |       | Riyan   |       |                       |
|                                                                  | Area                             | %age  | Area    | %age  | Area    | %age  | Area          | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area      | %age  | Area    | %age  |                       |
| < 1.5                                                            | 98.8                             | 6.3   | 436.0   | 26.6  | 358.4   | 16.3  | 1,220.3       | 75.1  | 460.2   | 36.1  | 20.9    | 2.0   | 383.7   | 28.4  | 89.3    | 3.8   | 306.0   | 12.0  | 315.8     | 30.7  | 78.8    | 7.0   | 3,768.2               |
| 1.5-3.0                                                          | 844.1                            | 53.5  | 1,094.1 | 66.6  | 1,528.1 | 69.4  | 338.1         | 20.8  | 446.2   | 35.0  | 473.5   | 44.0  | 958.2   | 71.0  | 1,508.9 | 64.6  | 1,115.6 | 43.7  | 547.7     | 53.4  | 778.1   | 70.0  | 9,632.6               |
| > 3.0                                                            | 635.7                            | 40.2  | 111.7   | 6.8   | 314.7   | 14.3  | 67.0          | 4.1   | 368.9   | 28.9  | 580.0   | 54.0  | 7.7     | 0.6   | 737.3   | 31.6  | 1,133.3 | 44.3  | 163.7     | 15.9  | 249.8   | 23.0  | 4,369.8               |
| Total                                                            | 1,578.6                          | 100.0 | 1,641.8 | 100.0 | 2,201.2 | 100.0 | 1,625.4       | 100.0 | 1,275.3 | 100.0 | 1,074.4 | 100.0 | 1,349.6 | 100.0 | 2,335.5 | 100.0 | 2,554.9 | 100.0 | 1,027.2   | 100.0 | 1,106.7 | 100.0 | 17,770.6              |

## GROUND WATER NITRATE DISTRIBUTION

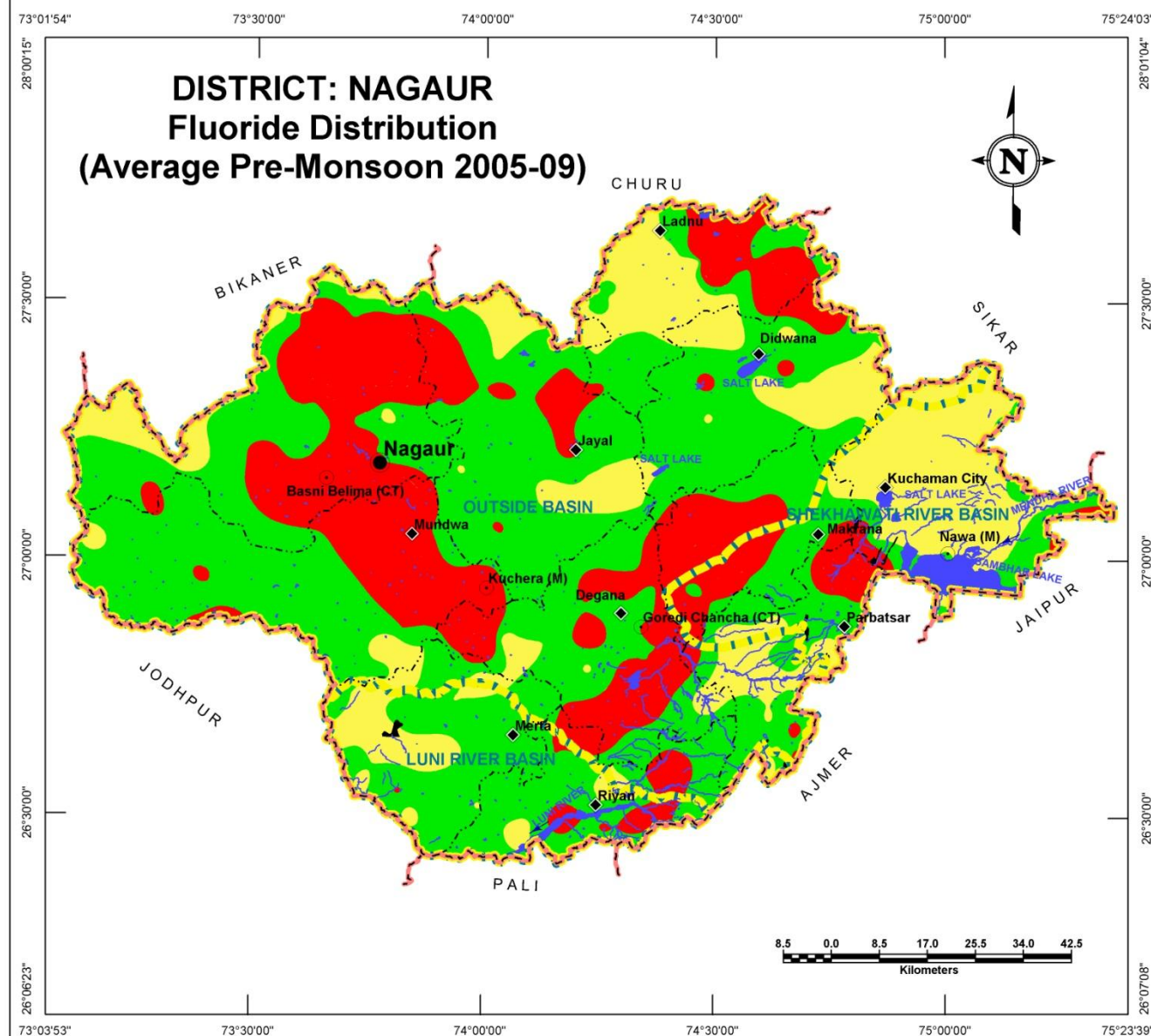
High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water of Nagaur district. High nitrate concentration (>100 mg/l) is shown in red color and occupies approximately 64% of the district area which is not suitable for agriculture purpose. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green color and occupy approximately 27% of the district area. Remaining part of the district has low nitrate concentration (<50 mg/l) in ground water, shown in yellow color. Such low Nitrate areas are largely seen around Metra, Parbatsar and Kuchman City where the ground water is suitable for agriculture purpose.

**Table: Block wise area of Nitrate distribution**

| Nitrate concentration<br>range(mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |         |       |               |       |         |       |         |       |         |       |         |       |         |       |           |       |         |       | Total<br>Area<br>(sq km) |
|-----------------------------------------------------------------|----------------------------------|-------|---------|-------|---------|-------|---------------|-------|---------|-------|---------|-------|---------|-------|---------|-------|---------|-------|-----------|-------|---------|-------|--------------------------|
|                                                                 | Degana                           |       | Didwana |       | Jayal   |       | Kuchaman City |       | Ladnu   |       | Makrana |       | Merta   |       | Mundwa  |       | Nagaur  |       | Parbatsar |       | Riyan   |       |                          |
|                                                                 | Area                             | %age  | Area    | %age  | Area    | %age  | Area          | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area      | %age  | Area    | %age  |                          |
| < 50                                                            | -                                | -     | 3.7     | 0.2   | 12.3    | 0.6   | 307.2         | 18.9  | 56.4    | 4.4   | 13.0    | 1.2   | 658.8   | 48.8  | 215.7   | 9.2   | 19.0    | 0.7   | 249.2     | 24.3  | 85.0    | 8.0   | 1,620.3                  |
| 50-100                                                          | 139.1                            | 8.8   | 66.1    | 4.1   | 310.7   | 14.1  | 461.0         | 28.4  | 319.0   | 25.0  | 88.4    | 8.2   | 607.8   | 45.0  | 934.5   | 40.0  | 683.6   | 26.8  | 561.1     | 54.6  | 655.9   | 59.0  | 4,827.2                  |
| >100                                                            | 1,439.5                          | 91.2  | 1,572.0 | 95.7  | 1,878.2 | 85.3  | 857.2         | 52.7  | 899.9   | 70.6  | 973.0   | 90.6  | 83.0    | 6.2   | 1,185.3 | 50.8  | 1,852.3 | 72.5  | 216.9     | 21.1  | 365.8   | 33.0  | 11,323.1                 |
| Total                                                           | 1,578.6                          | 100.0 | 1,641.8 | 100.0 | 2,201.2 | 100.0 | 1,625.4       | 100.0 | 1,275.3 | 100.0 | 1,074.4 | 100.0 | 1,349.6 | 100.0 | 2,335.5 | 100.0 | 2,554.9 | 100.0 | 1,027.2   | 100.0 | 1,106.7 | 100.0 | 17,770.6                 |



PLATE - XIV



**LEGEND**

**Admin Boundary:**

District Headquarter

Block Headquarter

Town

District Boundary

Block Boundary

**Basin Boundary**

**Water Bodies:**

River / Streams

Ponds / Reservoirs

Hills

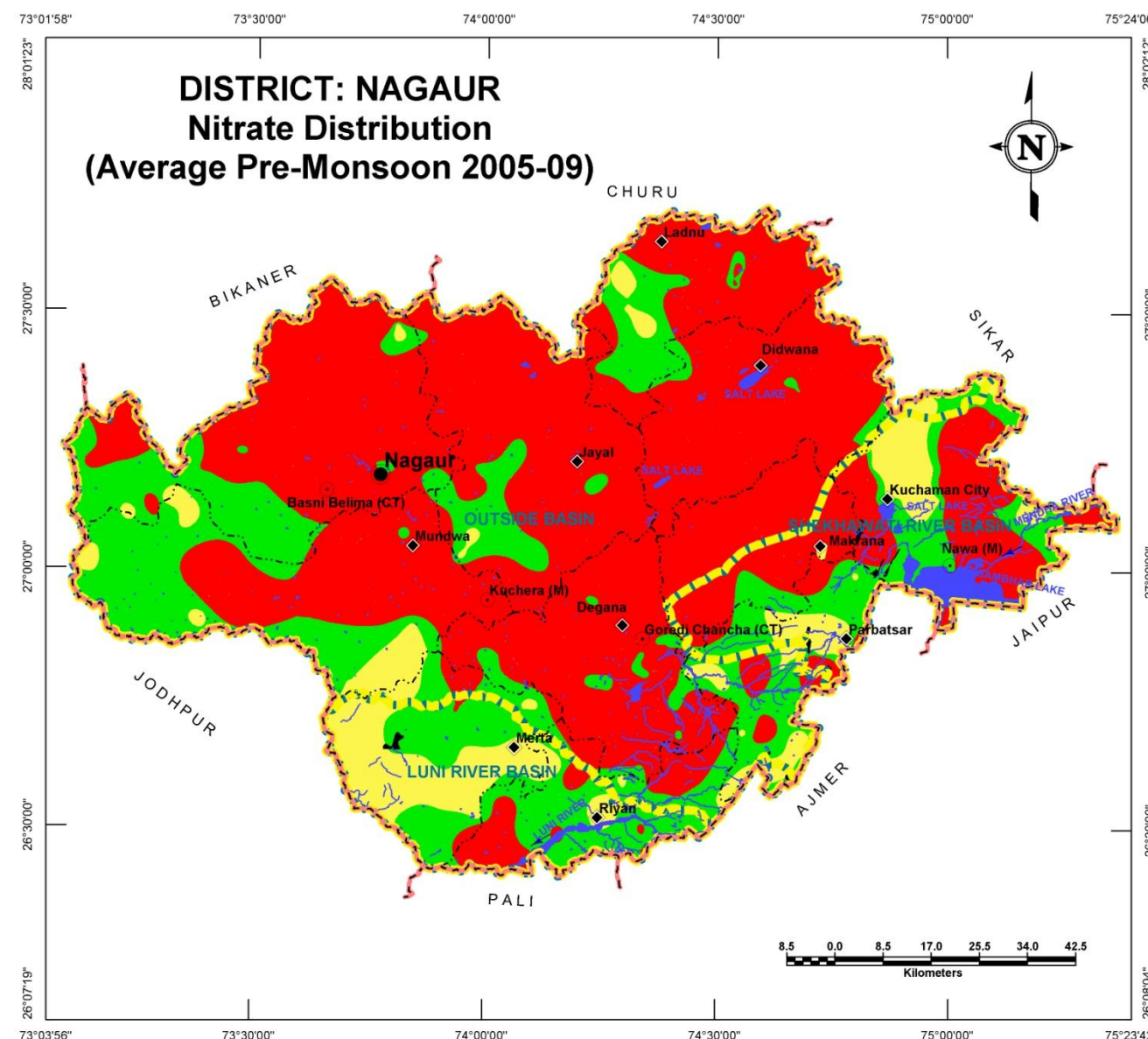
**Fluoride Concentration (mg/l):**

< 1.5

1.5-3.0

> 3.0

PLATE - XV



**LEGEND**

**Admin Boundary:**

District Headquarter

Block Headquarter

Town

District Boundary

Block Boundary

**Basin Boundary**

**Water Bodies:**

River / Streams

Ponds / Reservoirs

Hills

**Nitrate Concentration (mg/l):**

< 50

50-100

> 100

## DEPTH TO BEDROCK

## DISTRICT – NAGOUR

The thick alluvial deposits in Nagaur district are underlain by rock of different lithology and age. Plate – XVI depicts the distribution of bedrock depth below ground level. The beginning of massive bedrock has been considered for defining top of bedrock surface. The major rocks types occurring in the district are Limestone, Sandstones and Schist. These rocks are overlain by alluvial deposits of sand, clay, silt and admixture of these in different proportions and thicknesses. The map of depth to bedrock in meters below ground level reveals that the bedrock surface is highly undulating in the western part whereas it is relatively even northeast and southwards. It varies from less than 20m bgl to more than 260m bgl. Shallow bedrock depth (less than 20m bgl) is found in the north part of Landu block. Deepest occurrence of bedrock (indicating high alluvial thickness) is found in eastern part of Nagaur block (more than 260m bgl) covering a small area of eight sq km. The central, eastern and south parts of the district have the bedrock reaching upto 160m bgl whereas, Nagaur block has moderately deep bedrock i.e. in the range of 160m bgl to 260m bgl.

| Depth to bedrock<br>(m bgl) | Block wise area coverage (sq km) |       |         |       |         |       |               |       |         |       |         |       |         |       |         |       |         |       |           |       |         |       | Total Area<br>(sq km) |
|-----------------------------|----------------------------------|-------|---------|-------|---------|-------|---------------|-------|---------|-------|---------|-------|---------|-------|---------|-------|---------|-------|-----------|-------|---------|-------|-----------------------|
|                             | Degana                           |       | Didwana |       | Jayal   |       | Kuchaman City |       | Ladnu   |       | Makrana |       | Merta   |       | Mundwa  |       | Nagaur  |       | Parbatsar |       | Riyan   |       |                       |
|                             | Area                             | %age  | Area    | %age  | Area    | %age  | Area          | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area    | %age  | Area      | %age  | Area    | %age  |                       |
| < 20                        | -                                | -     | -       | -     | -       | -     | -             | -     | 26.4    | 2.0   | -       | -     | -       | -     | -       | -     | -       | -     | -         | -     | -       | -     | 26.4                  |
| 20-40                       | -                                | -     | -       | -     | -       | -     | -             | -     | 179.9   | 14.0  | -       | -     | -       | -     | 0.1     | -     | -       | -     | 193.8     | 19.0  | 19.6    | 2.0   | 393.4                 |
| 40-60                       | 6.4                              | -     | 98.2    | 6.0   | 169.0   | 7.7   | 610.7         | 37.6  | 160.5   | 13.0  | -       | -     | -       | -     | 1.0     | -     | 15.1    | 1.0   | 267.3     | 26.0  | 126.0   | 11.0  | 1,454.2               |
| 60-80                       | 25.0                             | 2.0   | 1,137.0 | 69.0  | 820.0   | 37.2  | 718.9         | 44.2  | 297.3   | 23.0  | 321.7   | 30.0  | -       | -     | 14.3    | 1.0   | 111.5   | 4.0   | 228.0     | 22.0  | 306.4   | 28.0  | 3,980.1               |
| 80-100                      | 1,112.8                          | 70.0  | 341.8   | 21.0  | 651.6   | 29.6  | 295.8         | 18.2  | 264.8   | 21.0  | 513.6   | 48.0  | 1,243.8 | 92.0  | 744.8   | 32.0  | 302.5   | 12.0  | 305.1     | 30.0  | 535.7   | 48.0  | 6,312.3               |
| 100-120                     | 347.5                            | 22.0  | 36.9    | 2.0   | 407.0   | 18.5  | -             | -     | 206.8   | 16.0  | 239.1   | 22.0  | 105.8   | 8.0   | 1,079.6 | 46.0  | 230.7   | 9.0   | 33.0      | 3.0   | 119.0   | 11.0  | 2,805.4               |
| 120-140                     | 86.9                             | 6.0   | 13.0    | 1.0   | 138.6   | 6.3   | -             | -     | 106.1   | 8.0   | -       | -     | -       | -     | 401.3   | 17.0  | 893.5   | 35.0  | -         | -     | -       | -     | 1,639.4               |
| 140-160                     | -                                | -     | 10.6    | 1.0   | 15.0    | 0.7   | -             | -     | 23.9    | 2.0   | -       | -     | -       | -     | 94.4    | 4.0   | 383.9   | 15.0  | -         | -     | -       | -     | 527.8                 |
| 160-180                     | -                                | -     | 4.3     | -     | -       | -     | -             | -     | 9.6     | 1.0   | -       | -     | -       | -     | -       | -     | 285.0   | 11.0  | -         | -     | -       | -     | 298.9                 |
| 180-200                     | -                                | -     | -       | -     | -       | -     | -             | -     | -       | -     | -       | -     | -       | -     | -       | -     | 173.7   | 7.0   | -         | -     | -       | -     | 173.7                 |
| 200-220                     | -                                | -     | -       | -     | -       | -     | -             | -     | -       | -     | -       | -     | -       | -     | -       | -     | 92.2    | 4.0   | -         | -     | -       | -     | 92.2                  |
| 220-240                     | -                                | -     | -       | -     | -       | -     | -             | -     | -       | -     | -       | -     | -       | -     | -       | -     | 35.4    | 1.0   | -         | -     | -       | -     | 35.4                  |
| 240-260                     | -                                | -     | -       | -     | -       | -     | -             | -     | -       | -     | -       | -     | -       | -     | -       | -     | 23.3    | 1.0   | -         | -     | -       | -     | 23.3                  |
| > 260                       | -                                | -     | -       | -     | -       | -     | -             | -     | -       | -     | -       | -     | -       | -     | -       | -     | 8.1     | -     | -         | -     | -       | -     | 8.1                   |
| Total                       | 1,578.6                          | 100.0 | 1,641.8 | 100.0 | 2,201.2 | 100.0 | 1,625.4       | 100.0 | 1,275.3 | 100.0 | 1,074.4 | 100.0 | 1,349.6 | 100.0 | 2,335.5 | 100.0 | 2,554.9 | 100.0 | 1,027.2   | 100.0 | 1,106.7 | 100.0 | 17,770.6              |

## UNCONFINED AQUIFER

### Alluvial areas:

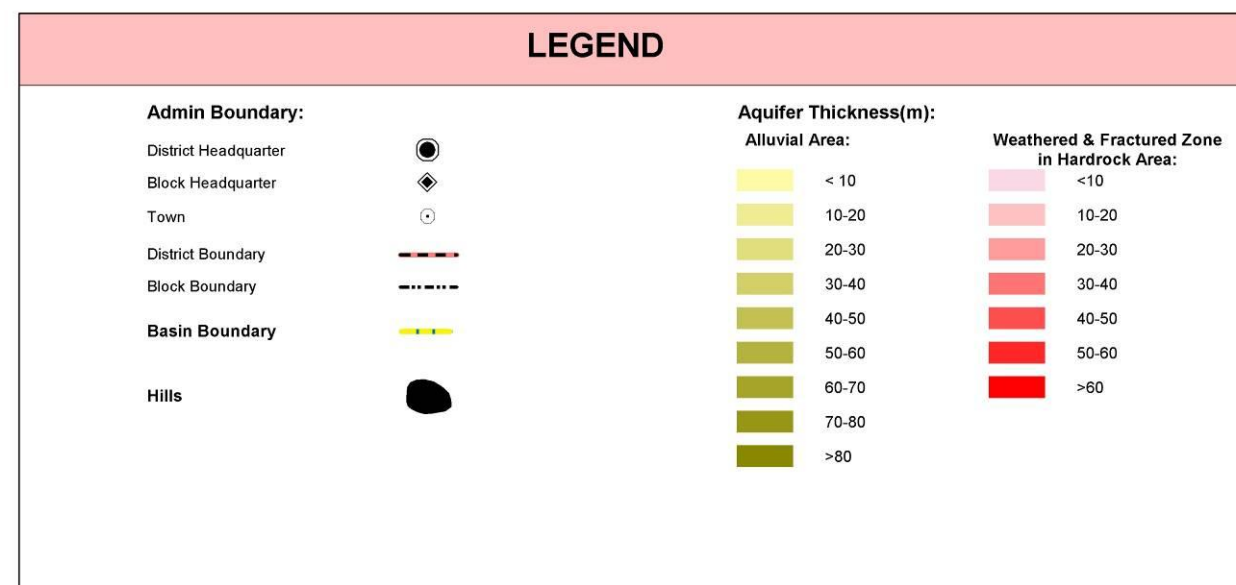
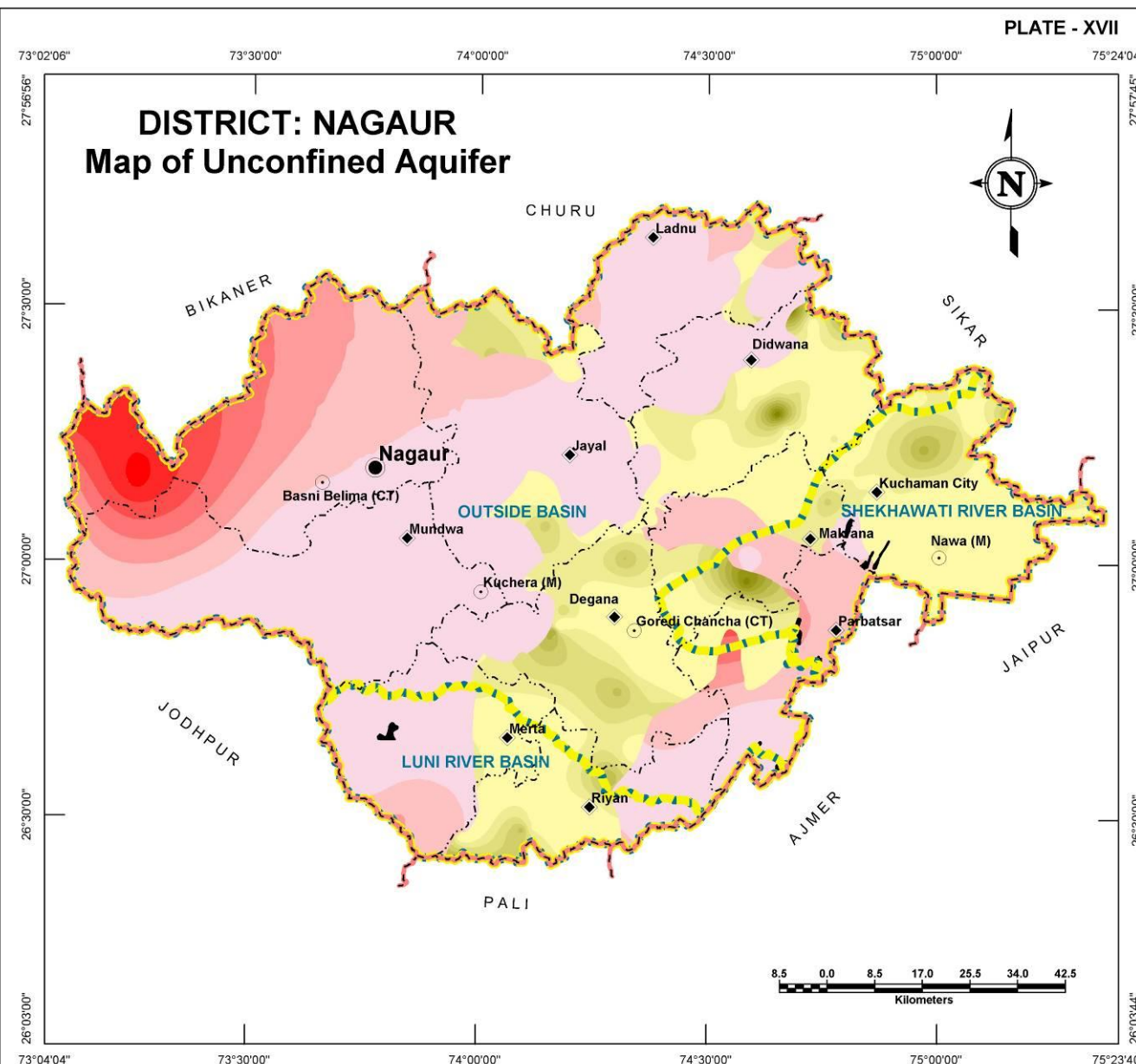
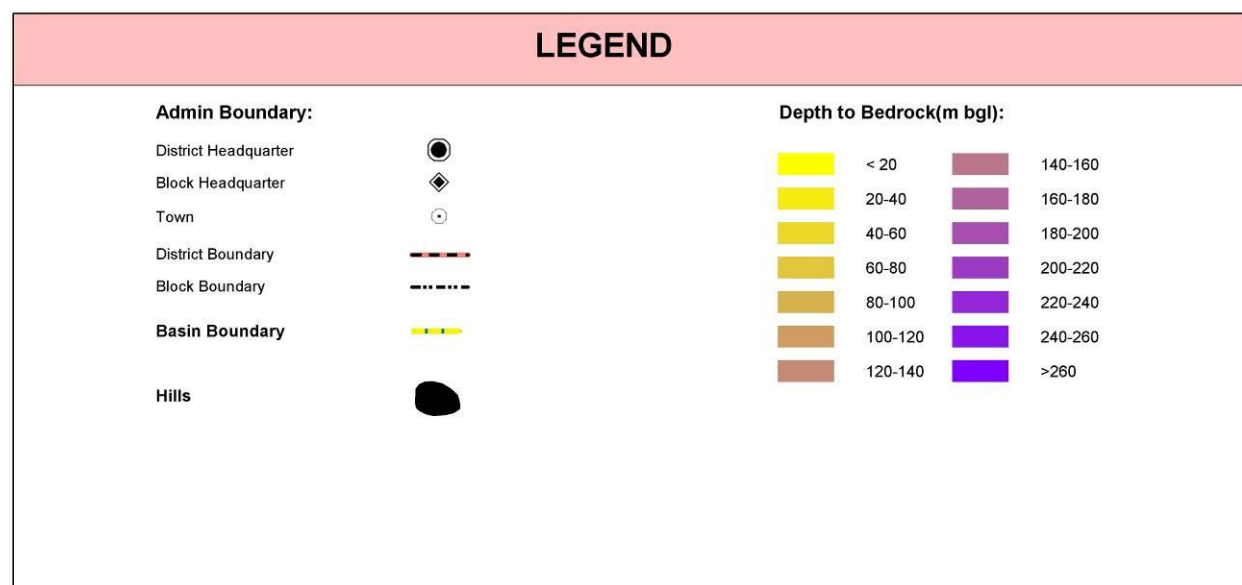
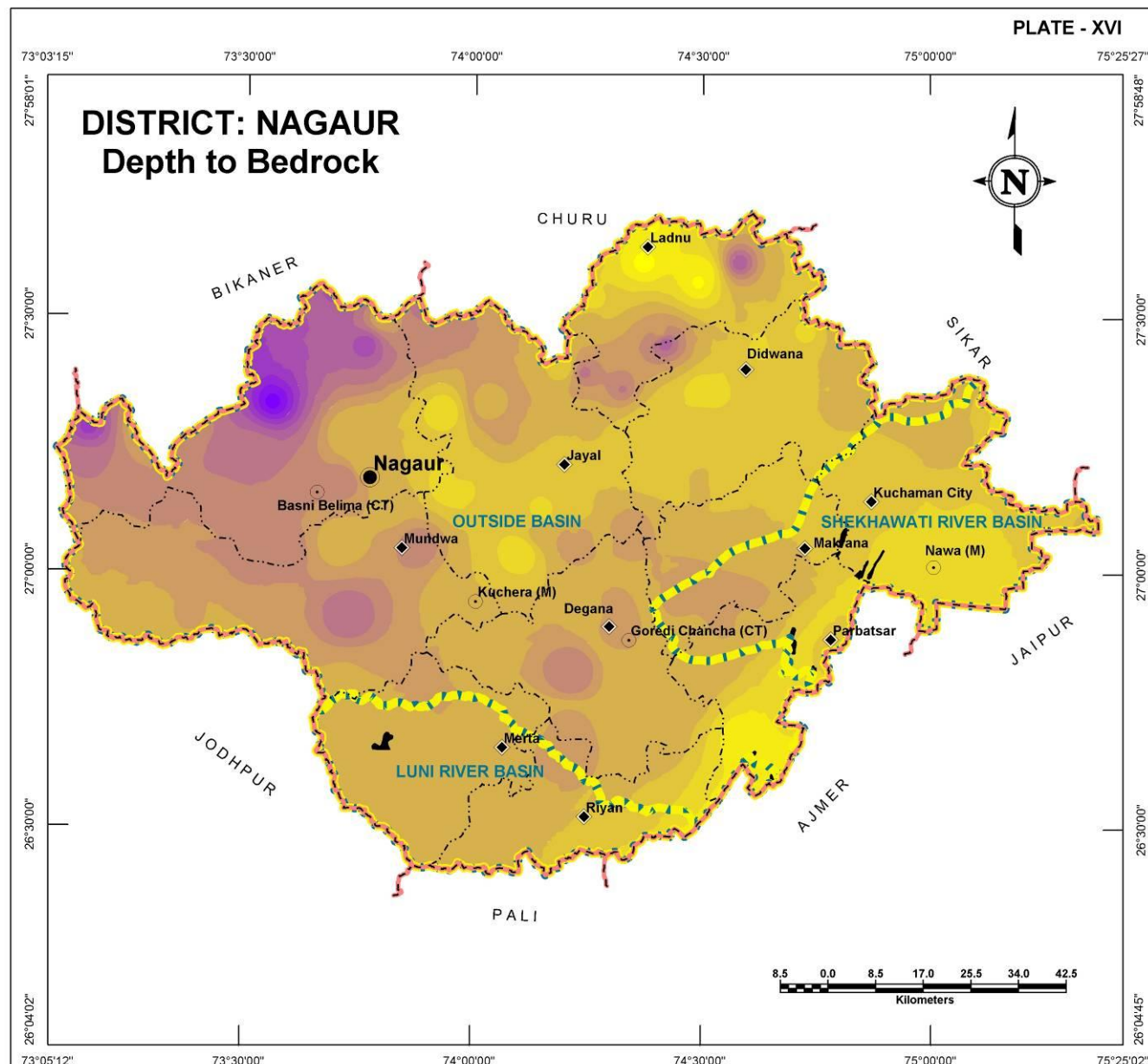
Alluvial material forms aquifers in most of northeastern and southeastern half of the district and also in the northern and southern parts of Jayal block. The thickness of unconfined aquifer varies from less than 10 m to about 90m with the thickest parts lying to the south of Makrana and the southeastern part of Didwana block. The general thickness of alluvial aquifers is however, upto 30m.

| Unconfined<br>aquifer<br>Thickness (m) | Block wise Area coverage(sq km) |                |              |                |              |              |              |          |          |              |              | Total<br>Area<br>(sq km) |
|----------------------------------------|---------------------------------|----------------|--------------|----------------|--------------|--------------|--------------|----------|----------|--------------|--------------|--------------------------|
|                                        | Degana                          | Didwana        | Jayal        | Kuchaman City  | Ladnu        | Makrana      | Merta        | Mundwa   | Nagaur   | Parbatsar    | Riyan        |                          |
| < 10                                   | 402.9                           | 740.0          | 206.7        | 1,120.1        | 43.0         | 371.1        | 238.9        | -        | -        | 259.0        | 442.0        | 3,823.7                  |
| 10-20                                  | 397.7                           | 259.2          | 162.5        | 267.8          | 75.9         | 171.2        | 69.2         | -        | -        | 38.3         | 111.7        | 1,553.5                  |
| 20-30                                  | 273.4                           | 60.3           | 137.7        | 100.5          | 2.4          | 61.0         | 0.1          | -        | -        | 15.6         | 62.6         | 713.6                    |
| 30-40                                  | 85.0                            | 31.5           | 8.0          | 64.2           | -            | 32.7         | -            | -        | -        | 4.5          | 66.9         | 292.8                    |
| 40-50                                  | 13.0                            | 20.1           | -            | 27.8           | -            | 23.7         | -            | -        | -        | 1.0          | 14.5         | 100.2                    |
| 50-60                                  | 3.0                             | 12.4           | -            | 3.9            | -            | 16.9         | -            | -        | -        | 0.2          | 2.3          | 38.7                     |
| 60-70                                  | -                               | 7.2            | -            | -              | -            | 12.7         | -            | -        | -        | -            | -            | 19.9                     |
| 70-80                                  | -                               | 4.3            | -            | -              | -            | 12.3         | -            | -        | -        | -            | -            | 16.6                     |
| > 80                                   | -                               | 1.6            | -            | -              | -            | 1.7          | -            | -        | -        | -            | -            | 3.3                      |
| <b>Total</b>                           | <b>1,175.0</b>                  | <b>1,136.6</b> | <b>514.9</b> | <b>1,584.3</b> | <b>121.3</b> | <b>703.3</b> | <b>308.2</b> | <b>-</b> | <b>-</b> | <b>318.6</b> | <b>700.0</b> | <b>6,562.2</b>           |

### Hard rock areas:

Weathered, fractured and jointed rock formations occurring at shallower depths constitute good unconfined aquifers. Such zone ranges in thickness from less than 10m to about 70m. Western part of the district in Nagaur block has thickest zone of weathered/fractured hardrock where a thickness of more than 60m is seen whereas the rest of the blocks has moderate thickness of upto 30m.

| Unconfined<br>aquifer<br>Thickness (m) | Block wise Area coverage (sq km) |              |                |               |                |              |                |                |                |              |              | Total<br>Area<br>(sq km) |
|----------------------------------------|----------------------------------|--------------|----------------|---------------|----------------|--------------|----------------|----------------|----------------|--------------|--------------|--------------------------|
|                                        | Degana                           | Didwana      | Jayal          | Kuchaman City | Ladnu          | Makrana      | Merta          | Mundwa         | Nagaur         | Parbatsar    | Riyan        |                          |
| <10                                    | 280.5                            | 447.0        | 1,287.9        | 34.7          | 1,010.4        | 43.1         | 823.4          | 1,734.7        | 418.4          | 240.8        | 377.0        | 6,698.1                  |
| 10-20                                  | 121.8                            | 58.2         | 398.4          | 6.4           | 143.4          | 322.6        | 209.2          | 273.6          | 775.2          | 381.9        | 28.4         | 2,719.2                  |
| 20-30                                  | 1.3                              | -            | -              | -             | 0.2            | 5.4          | 8.8            | 160.5          | 614.3          | 56.9         | 1.3          | 848.7                    |
| 30-40                                  | -                                | -            | -              | -             | -              | -            | -              | 113.1          | 264.9          | 25.6         | -            | 403.6                    |
| 40-50                                  | -                                | -            | -              | -             | -              | -            | -              | 52.9           | 188.0          | 3.4          | -            | 244.3                    |
| 50-60                                  | -                                | -            | -              | -             | -              | -            | -              | 0.7            | 262.8          | -            | -            | 263.5                    |
| > 60                                   | -                                | -            | -              | -             | -              | -            | -              | -              | 31.3           | -            | -            | 31.3                     |
| <b>Total</b>                           | <b>403.6</b>                     | <b>505.2</b> | <b>1,686.3</b> | <b>41.1</b>   | <b>1,154.0</b> | <b>371.1</b> | <b>1,041.4</b> | <b>2,335.5</b> | <b>2,554.9</b> | <b>708.6</b> | <b>406.7</b> | <b>11,208.4</b>          |



## Glossary of terms

| S. No. | Technical Terms        | Definition                                                                                                                                   |
|--------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | AQUIFER                | A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.          |
| 2      | ARID CLIMATE           | Climate characterized by high evaporation and low precipitation.                                                                             |
| 3      | ARTIFICIAL RECHARGE    | Addition of water to a ground water reservoir by man-made activity                                                                           |
| 4      | CLIMATE                | The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region. |
| 5      | CONFINED AQUIFER       | A water bearing strata having confined impermeable overburden. In this aquifer, water levels represent the piezometric head.                 |
| 6      | CONTAMINATION          | Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.                      |
| 7      | DRAWDOWN               | The drawdown is the depth by which water level is lowered.                                                                                   |
| 8      | FRESH WATER            | Water suitable for drinking purpose.                                                                                                         |
| 9      | GROUND WATER           | Water found below the land surface.                                                                                                          |
| 10     | GROUND WATER BASIN     | A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.                                           |
| 11     | GROUND WATER RECHARGE  | The natural infiltration of surface water into the ground.                                                                                   |
| 12     | HARD WATER             | The water which does not produce sufficient foam with soap.                                                                                  |
| 13     | HYDRAULIC CONDUCTIVITY | A constant that serves as a measure of permeability of porous medium.                                                                        |
| 14     | HYDROGEOLOGY           | The science related with the ground water.                                                                                                   |
| 15     | HUMID CLIMATE          | The area having high moisture content.                                                                                                       |
| 16     | ISOHYET                | A line of equal amount of rainfall.                                                                                                          |
| 17     | METEOROLOGY            | Science of the atmosphere.                                                                                                                   |
| 18     | PERCOLATION            | It is flow through a porous substance.                                                                                                       |
| 19     | PERMEABILITY           | The property or capacity of a soil or rock for transmitting water.                                                                           |
| 20     | pH                     | Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).                               |
| 21     | PIEZOMETRIC HEAD       | Elevation to which water will rise in a piezometers.                                                                                         |
| 22     | RECHARGE               | It is a natural or artificial process by which water is added from outside to the aquifer.                                                   |
| 23     | SAFE YIELD             | Amount of water which can be extracted from ground water without producing undesirable effect.                                               |
| 24     | SALINITY               | Concentration of dissolved salts.                                                                                                            |
| 25     | SEMI-ARID              | An area is considered semiarid having annual rainfall between 10-20 inches.                                                                  |
| 26     | SEMI-CONFINED AQUIFER  | Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.                                                                  |
| 27     | SPECIFIC YIELD         | Quantity of water which is released by a formation after it's complete saturation.                                                           |
| 28     | TOTAL DISSOLVED SOLIDS | Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.                                  |

(Contd...)

| S. No. | Technical Terms            | Definition                                                                                                                                                                                                                   |
|--------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29     | TRANSMISSIBILITY           | It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability. |
| 30     | UNCONFINED AQUIFER         | A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.                                                                                                              |
| 31     | UNSATURATED ZONE           | The zone below the land surface in which pore space contains both water and air.                                                                                                                                             |
| 32     | WATER CONSERVATION         | Optimal use and proper storage of water.                                                                                                                                                                                     |
| 33     | WATER RESOURCES            | Availability of surface and ground water.                                                                                                                                                                                    |
| 34     | WATER RESOURCES MANAGEMENT | Planned development, distribution and use of water resources.                                                                                                                                                                |
| 35     | WATER TABLE                | Water table is the upper surface of the zone of saturation at atmospheric pressure.                                                                                                                                          |
| 36     | ZONE OF SATURATION         | The ground in which all pores are completely filled with water.                                                                                                                                                              |
| 37     | ELECTRICAL CONDUCTIVITY    | Flow of free ions in the water at 25C $\mu\text{m}/\text{cm}$ .                                                                                                                                                              |
| 38     | CROSS SECTION              | A Vertical Projection showing sub-surface formations encountered in a specific plane.                                                                                                                                        |
| 39     | 3-D PICTURE                | A structure showing all three dimensions i.e. length, width and depth.                                                                                                                                                       |
| 40     | GWD                        | Ground Water Department                                                                                                                                                                                                      |
| 41     | CGWB                       | Central Ground Water Board                                                                                                                                                                                                   |
| 42     | CGWA                       | Central Ground Water Authority                                                                                                                                                                                               |
| 43     | SWRPD                      | State Water Resources Planning Department                                                                                                                                                                                    |
| 44     | EU-SPP                     | European Union State Partnership Programme                                                                                                                                                                                   |
| 45     | TOPOGRAPHY                 | Details of drainage lines and physical features of land surface on a map.                                                                                                                                                    |
| 46     | GEOLOGY                    | The science related with the Earth.                                                                                                                                                                                          |
| 47     | GEOMORPHOLOGY              | The description and interpretation of land forms.                                                                                                                                                                            |
| 48     | PRE MONSOON SURVEY         | Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)                                                                                                 |
| 49     | POST-MONSOON SURVEY        | Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)                                                                                          |
| 50     | PIEZOMETER                 | A non-pumping small diameter bore hole used for monitoring of static water level.                                                                                                                                            |
| 51     | GROUND WATER FLUCTUATION   | Change in static water level below ground level.                                                                                                                                                                             |
| 52     | WATER TABLE                | The static water level found in unconfined aquifer.                                                                                                                                                                          |
| 53     | DEPTH OF BED ROCK          | Hard & compact rock encountered below land Surface.                                                                                                                                                                          |
| 54     | G.W. MONITORING STATION    | Dug wells selected on grid basis for monitoring of state water level.                                                                                                                                                        |
| 55     | EOLIAN DEPOSITS            | Wind-blown sand deposits                                                                                                                                                                                                     |



### बोरोन-स्नायु तन्त्र पर प्रभाव

क्लोराइड-सोडियम के साथ  
मिलकर उच्च रक्त चाप

### कैल्शियम-जोड़ों में कड़ापन

आर्सेनिक-त्वचा रोग, कैंसर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाध वयस्कों में गर्दे के रोग

फ्लोराइड - दंत क्षरण

सोडियम-हृदय, गुर्दा व रक्त  
परिसंचरण रोगों से  
ग्रसित लोगों को हानिकारक

नाइट्रेट-नवजात शिशुओं  
ब्लू बेबी बीमारी  
(मेथेमोग्लोबिनिमिया)

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

आयरन-आयरन जीवाणु से  
आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न,  
हड्डियों में मड़ाव



**भूजल अमूल्य है इसे प्रदूषित न करें।**

## Myths and Facts about Ground Water

| S No | Myths                                                                                                                                                                        | Facts                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | What is Ground Water <ul style="list-style-type: none"> <li>• an underground lake</li> <li>• a net work of underground rivers</li> <li>• a bowl filled with water</li> </ul> | Water which occurs below the land in geological formations/rocks is Ground water                                                                                                       |
| 2    | Ground Water occurs everywhere beneath the Land Surface                                                                                                                      | Not really, it depends on the nature of rock formation                                                                                                                                 |
| 3    | There is a relationship between ground water and surface water                                                                                                               | Not all the places. Near streams/rivers there is relation                                                                                                                              |
| 4    | Groundwater is not renewable resource                                                                                                                                        | It is renewable source and every year it is being recharged through rain/applied irrigation etc                                                                                        |
| 5    | Ground water is unlimited and deeper you drill more discharge                                                                                                                | It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper                                                                         |
| 6    | Ground Water moves rapidly                                                                                                                                                   | The movement of ground water is very slow                                                                                                                                              |
| 7    | Ground water pumped from wells is thousands of years old                                                                                                                     | Generally the ground water being tapped through wells is a few years old                                                                                                               |
| 8    | If water taste good—it is safe to drink                                                                                                                                      | It may have other chemicals e.g. fluoride, nitrates etc which are harmful                                                                                                              |
| 9    | Water from free flowing tube wells is very pure                                                                                                                              | This water can also be contaminated so test before use                                                                                                                                 |
| 10   | If I recharge my TW/DW/HP it will not benefit me                                                                                                                             | It will also benefit you and also adjoining wells                                                                                                                                      |
| 11   | There is no static ground water resources in Rajasthan                                                                                                                       | Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge                                                   |
| 12   | I cannot meet annual cooking and drinking water requirement by rain water harvesting                                                                                         | The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement |
| 13   | You can increase ground water recharge                                                                                                                                       | This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures                                                        |
| 14   | You cannot use abandoned TW/HP/DW for ground water recharge                                                                                                                  | These should be used as recharge structures as harvested rain water is directly put into GW reservoir                                                                                  |
| 15   | Putting waste near HP/TW will not cause any problem                                                                                                                          | Such actions will pollute wells and water                                                                                                                                              |



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